

CLIENT USER MANUAL

ARAPI

Complete Client Guide

Import. Design. Execute. Export.

Version 1.0.0 | Windows Desktop

Reviewed 26 August 2026

Document control

Field	Value
Product	ARAPI Windows desktop application
Guide version	1.0.0
Audience	Banking operations staff, QA analysts, implementation teams, and administrators
Scope	ARAPI only; AR Conversational is a separate application and is not documented here
Evidence basis	ARAPI 1.0.0 source, running desktop build, and the screenshots reproduced in this guide
Review date	26 August 2026

Client-safe scope: The examples use mock or placeholder values. Never copy production API keys, access tokens, customer data, or confidential response bodies into documentation or support tickets.

How to use this guide

- Start with Sections 2-4 before operating a new installation.
- Use Section 6 as the visual, screen-by-screen walkthrough; every screenshot is from the running ARAPI desktop app.
- Use Section 7 when configuring Designer commands and Section 9 when generating or running Bash/curl files.
- Use Appendix A to look up every visible control and Appendix B only for approved local integration/diagnostic work.

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1. About ARAPI

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ARAPI is a Windows desktop application for catalog-driven API testing. It imports real OpenAPI/Swagger specifications, presents their endpoints in a banking taxonomy, lets users assemble ordered BotJobs, and exports Bash/curl files for command-line execution.

The user interface is local and communicates with a localhost C# backend. No login is required inside the desktop app; workstation access, file permissions, network controls, and upstream API authentication remain the customer's responsibility.

Core concepts

Catalog: The imported, searchable list of real API methods and paths.

BotJob: A saved ordered workflow made of commands and variables.

Environment: A named target base URL plus optional default headers.

Run: A saved local execution/audit record with step rows.

Bash export: A generated .sh file containing enabled API_CALL commands as curl invocations.

Recommended operating sequence

1. Import specifications and confirm the catalog.
2. Create an approved environment or use the built-in mock context.
3. Create/review the BotJob and Save it.
4. Generate a Bash script into a controlled directory and review it.
5. Set required variables, run the script, and retain target-side evidence.
6. Use Execute Tests and Reports for the ARAPI-local audit and exports.

2. Read this before using the current release

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Version basis: This manual describes the observed ARAPI 1.0.0 desktop build as reviewed on 26 August 2026. Screen counts and demo data will differ by workstation.

Current-build qualifications

- Test Cases is a reserved workspace: New test case is disabled in version 1.0.0.

- IF, ELSE, LOOP, FOR_EACH, READ_CSV, READ_EXCEL, AI_GENERATE_DATA, and CALL_COMPONENT are preview palette items. The Designer states that they are not executed and are recorded as skipped.
- BotJob-specific Bash generation includes enabled API_CALL commands only. Assertions, extraction, waits, and control/data commands are not emitted into the .sh file.
- Bot Builder may create a synthetic, mock-only endpoint/data record when no imported endpoint matches strongly enough. Review its evidence and never treat a synthetic record as an imported production contract.
- The observed desktop C# build uses the Execute Tests page as a local run/audit view. A green row is not independent proof that a real downstream banking system changed state; use the Bash execution path plus target-side logs/data for acceptance.
- Mock Start/Stop controls the in-app state/log in the observed build. Do not assume port 8855 is an externally callable HTTP listener unless the deployment administrator has enabled and verified one.
- The AI provider Test control is a local configuration indicator in the observed build. Confirm provider connectivity with a real Bot Builder request.

Production safeguard: Before pointing any environment at a non-mock system, obtain authorization, review the generated curl file, restrict credentials to least privilege, and agree on expected side effects and rollback.

3. Installation and first launch

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Install or unpack

1. Use the ARAPI installer supplied by your organization, or unpack the approved portable ARAPI bundle into a user-writable folder.
2. Keep the executable, packaged backend, and any supplied data directory together. Do not move individual sidecar files out of the distribution.
3. Launch ARAPI. The app opens a single resizable desktop window; no web login is required.

Verify first launch

- Confirm the top-right indicator becomes sidecar online.
- Open Home and confirm Engine status is online.
- Open Settings and confirm Sidecar port 8787 is displayed.

- If a portable distribution includes a data folder next to ARAPI.exe, preserve and back up that folder before upgrades. Other distributions use an application-data location selected by the desktop shell.

Local-only service: The backend is intended for localhost use. Do not publish port 8787 or 8855 on a public network interface.

4. End-to-end quick start

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1. Confirm sidecar online on Home.
2. Open Import APIs. Use Upload Files for selected files/folders, or Folder (Desktop) for an existing local specification directory. Import .yaml, .yml, or .json files.
3. Open API Catalog. Search for a known method/path and confirm the summary and mapping status.
4. Open Environments. For a real approved target, create a named environment with its base URL and non-secret defaults. Keep credentials in variables where possible.
5. Open BotJob Designer. Create/select a BotJob, add variables, add and configure commands, order them, and Save.
6. Open Scripts (or select the BotJob in Execute Tests and choose Create Scripts). Select the environment and Browse to Documents\ARAPI\Scripts.
7. Select Create Bash Script. Review the timestamped .sh file; set every required environment variable; run with Bash and curl.
8. Use Execute Tests for the ARAPI-local run audit, then Reports for HTML/CSV or whole-catalog Postman/Bash downloads.

Best output directory: Use a dedicated user-writable directory such as %USERPROFILE%\Documents\ARAPI\Scripts. Do not save generated scripts inside the application installation folder.

5. Interface fundamentals

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Global header

Page name: The left side of the top bar repeats the current workspace.

Employee / Client: Changes shared conversation-mode context where a mode-aware experience is used. It does not change catalog contents, environment definitions, or BotJob command order.

sidecar status: Green/online means the React interface can reach the local backend. Resolve offline before importing, saving, or running.

Sidebar

Destination	Purpose
Home	Overview, engine status, and shortcuts to every ARAPI area.
Import APIs	Load OpenAPI or Swagger documents from files or a folder.
API Catalog	Search the imported method/path inventory and mapping state.
Business Categories	Browse the 25-domain banking taxonomy.
Test Cases	Reserved reusable-test workspace; creation is disabled in this release.
Bot Builder	Use conversational actions to search, create BotJobs, and request runs.
BotJob Designer	Create, configure, enable/disable, and order commands.
Environments	Manage named base URLs and default headers.
Scripts	Generate a BotJob-specific Bash file containing ordered curl calls.
Execute Tests	Select a BotJob/environment and review the local run audit.
Mock Server	Control the in-app mock state and inspect its request/system log.
Reports	Download run reports and whole-catalog exports.
Settings	Configure AI provider records and view local-service ports.

Common visual states

- Blue primary button: the main action for the page.
- Dimmed button or palette item: required selection/data is missing, or the feature is reserved.
- Green badge: passed, active default, running, or online depending on context.
- Orange/red text: warning, failure, missing mapping, or validation error that requires review.
- Chevron: expand/collapse a provider, command, action card, or result row.

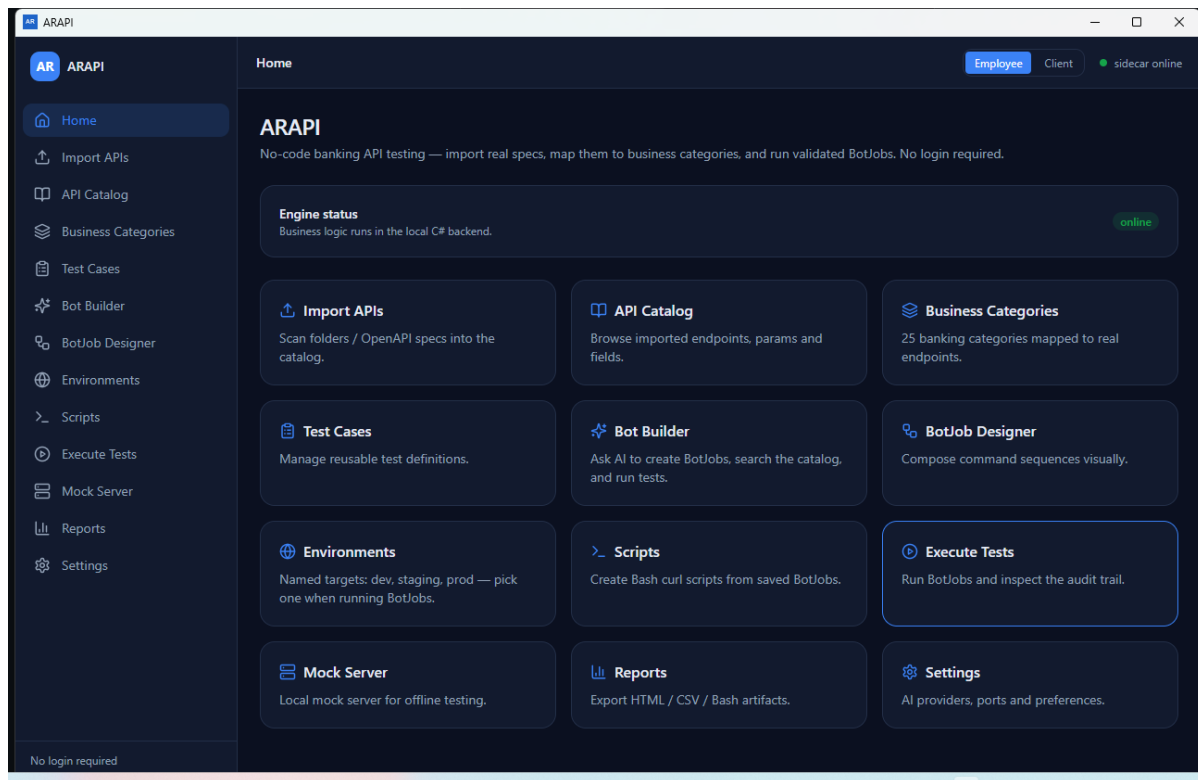


Figure 1. Home overview

Home provides engine health and shortcuts. Use any card or the matching sidebar item to open a workspace.

6. Screen-by-screen guide

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The following pages show every main ARAPI screen and the important interaction states captured from the running desktop application. Values are demonstration data; workflows and controls are the focus.

6.2 Import APIs - Folder (Desktop)

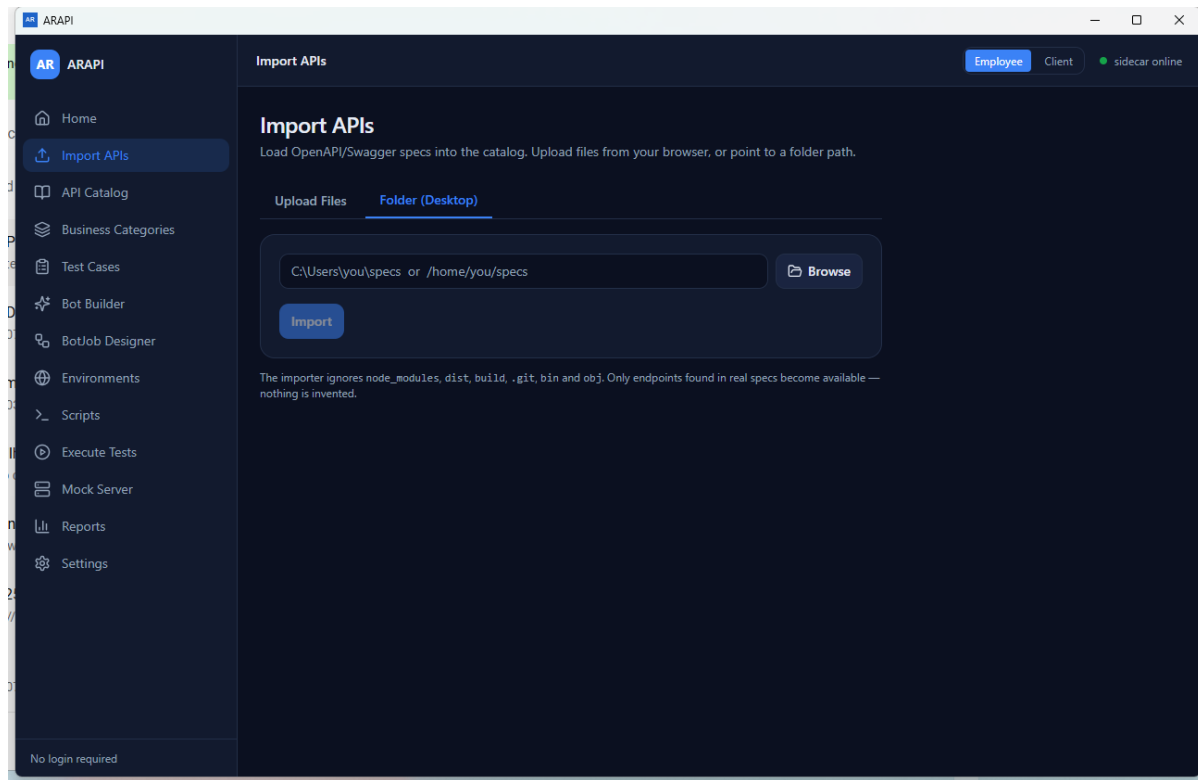


Figure 2. Import APIs - Folder (Desktop)

Choose Browse, select a specification folder, confirm the path, and select Import. Subfolders are scanned.

Importer scope: The recursive scan ignores node_modules, dist, build, .git, bin, and obj. Only endpoints present in parsed specifications enter the catalog.

6.3 Import APIs - Upload Files

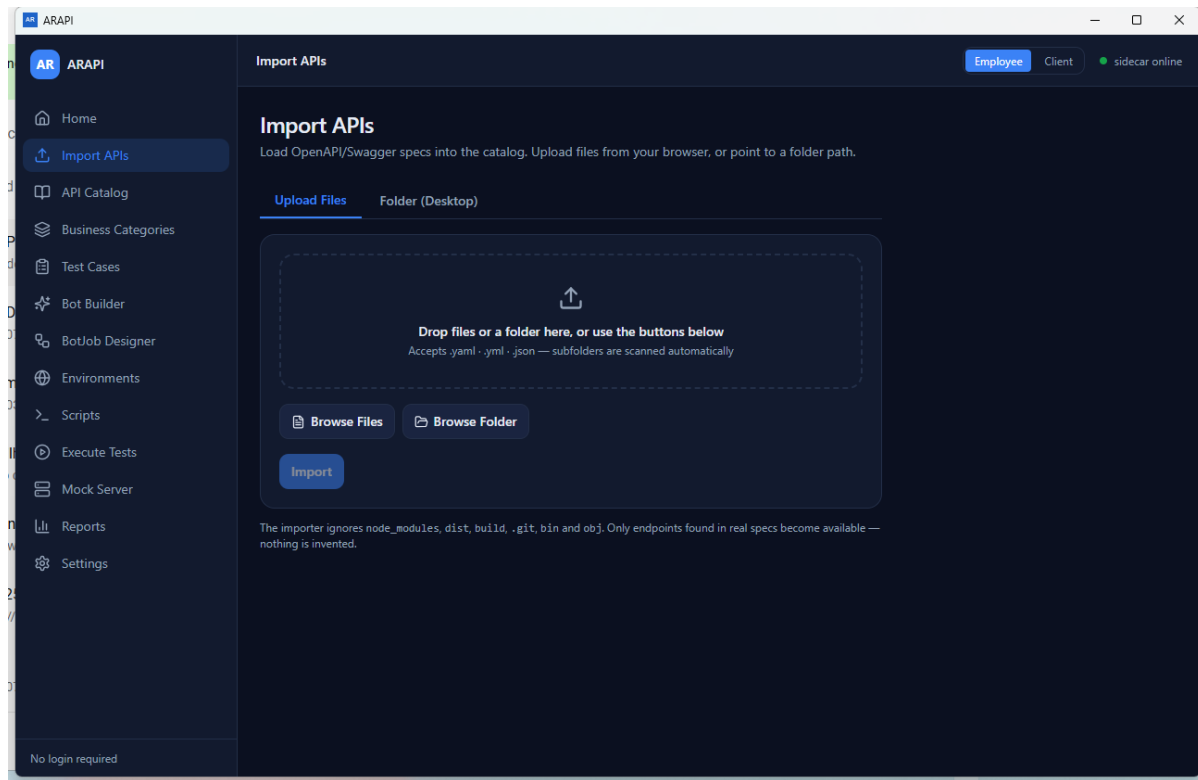


Figure 3. Import APIs - Upload Files

Drop YAML/JSON files or a folder, or use the browse buttons. Remove unwanted items before selecting Import.

Supported files: .yaml, .yml, and .json. Review the import summary and each reported failure before continuing.

6.4 API Catalog

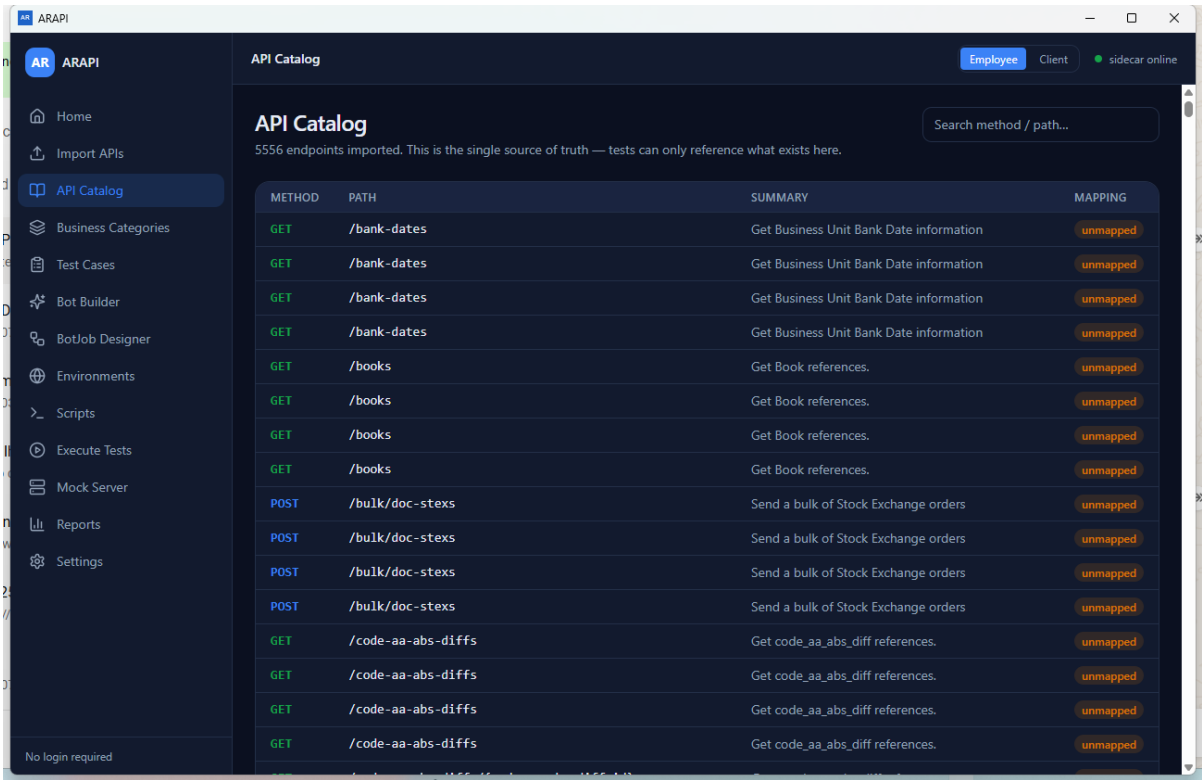


Figure 4. API Catalog

Search by HTTP method, path, or summary. Mapped/unmapped indicates whether taxonomy mapping exists.

Single source of selection: Designer API_CALL controls can select only endpoint IDs currently returned by the catalog.

6.5 Business Categories

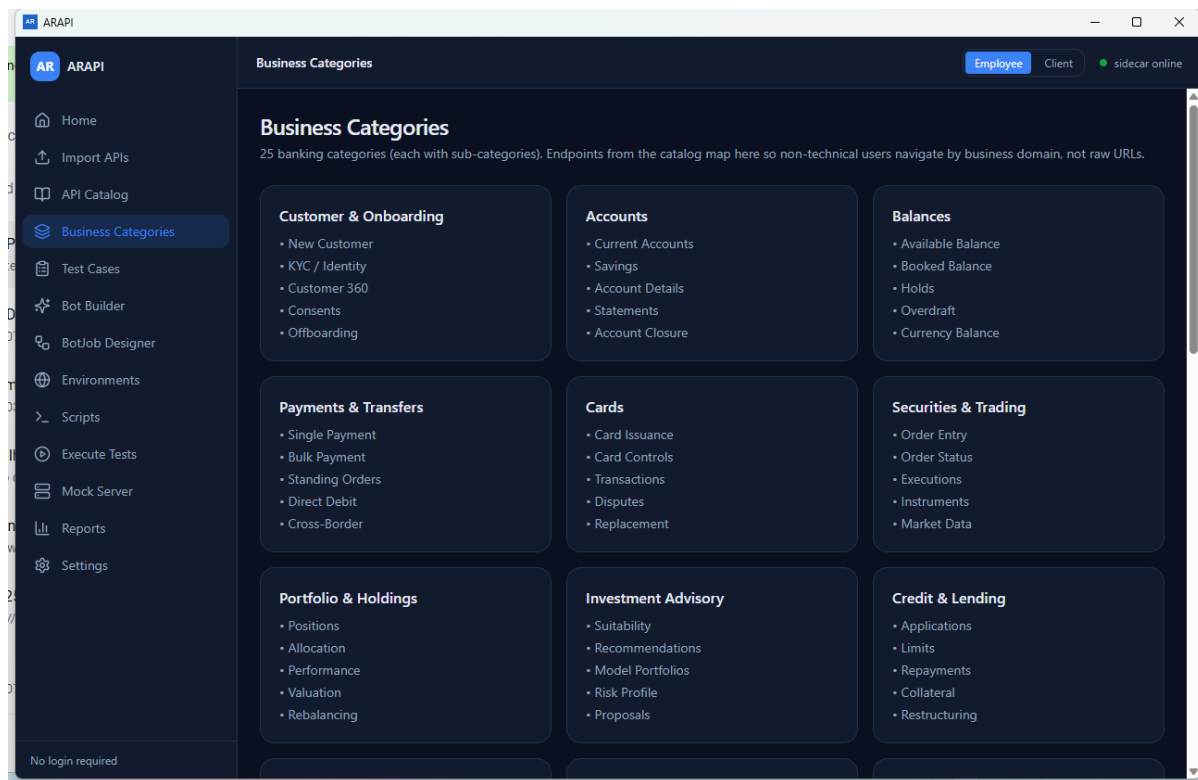


Figure 5. Business Categories

This is a read-only business view of the 25-domain taxonomy in the current UI.

6.6 Test Cases

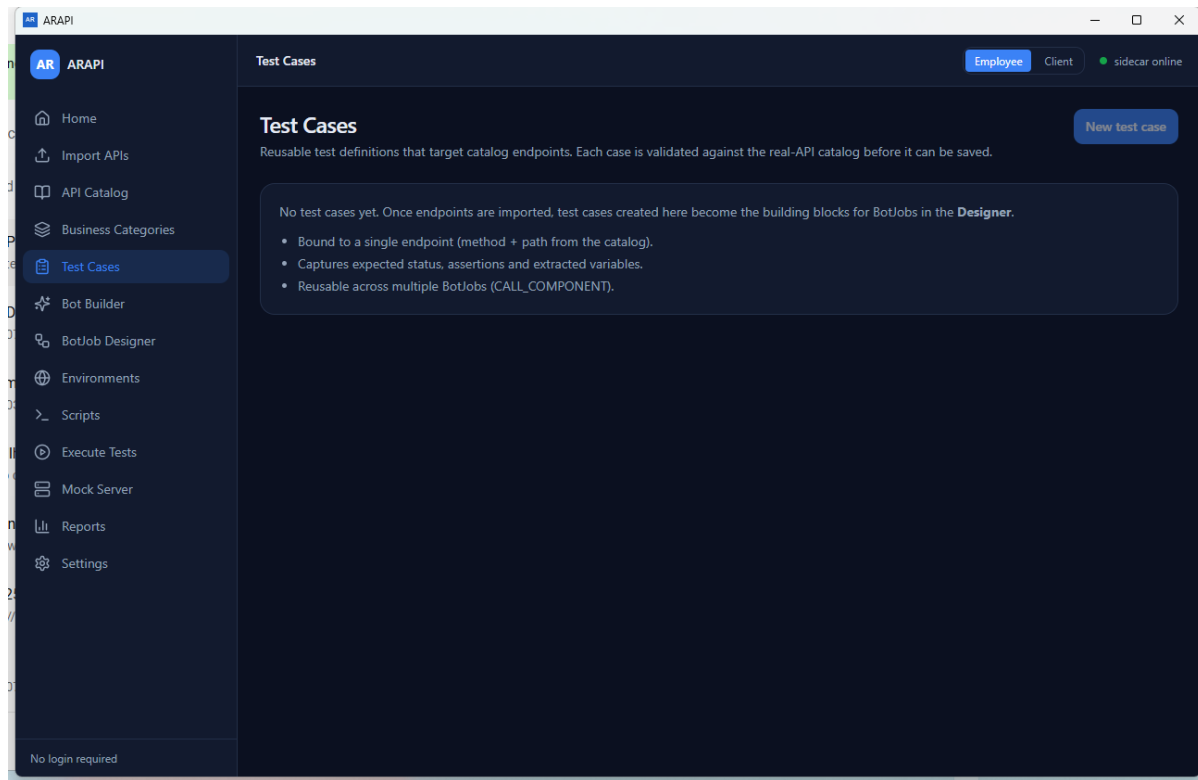


Figure 6. Test Cases

The page documents the intended reusable-test concept, but New test case is disabled in version 1.0.0.

Release status: New test case is disabled in 1.0.0. Use BotJob Designer for available workflow authoring.

6.7 Bot Builder - start

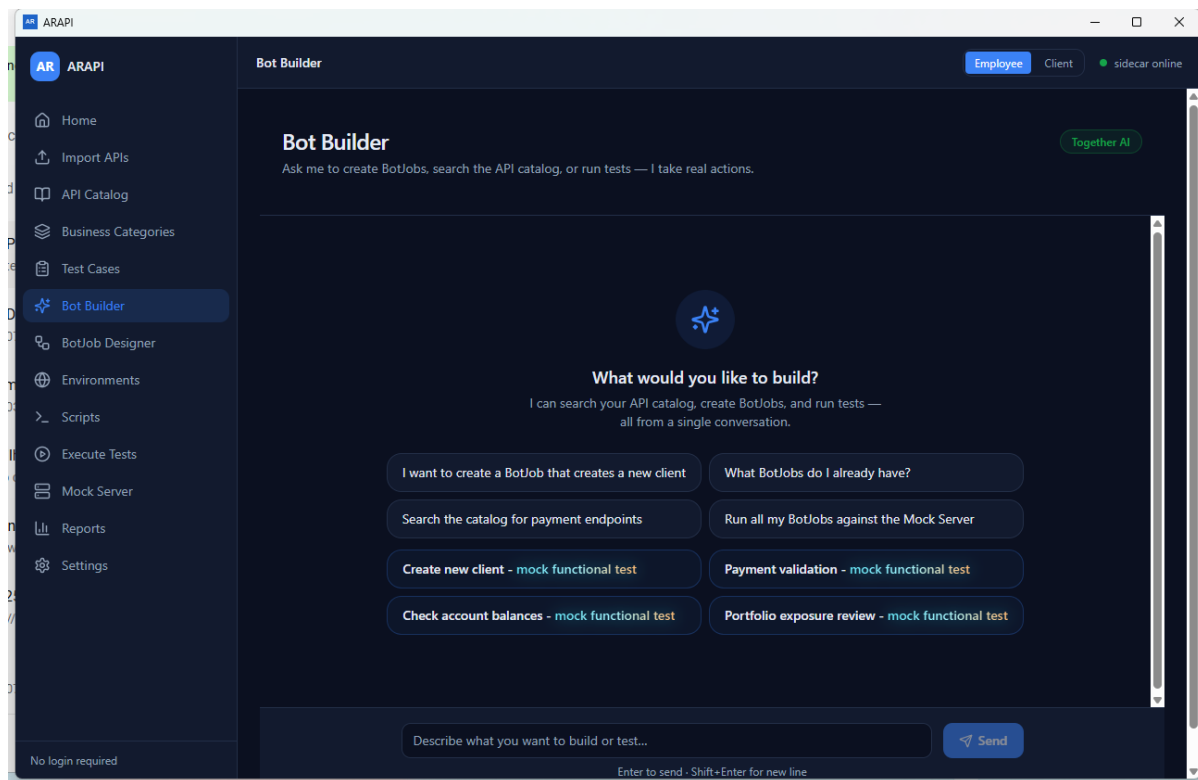


Figure 7. Bot Builder - start

Select a suggestion or write an instruction. Enter sends; Shift+Enter inserts a new line.

6.8 Bot Builder - catalog action

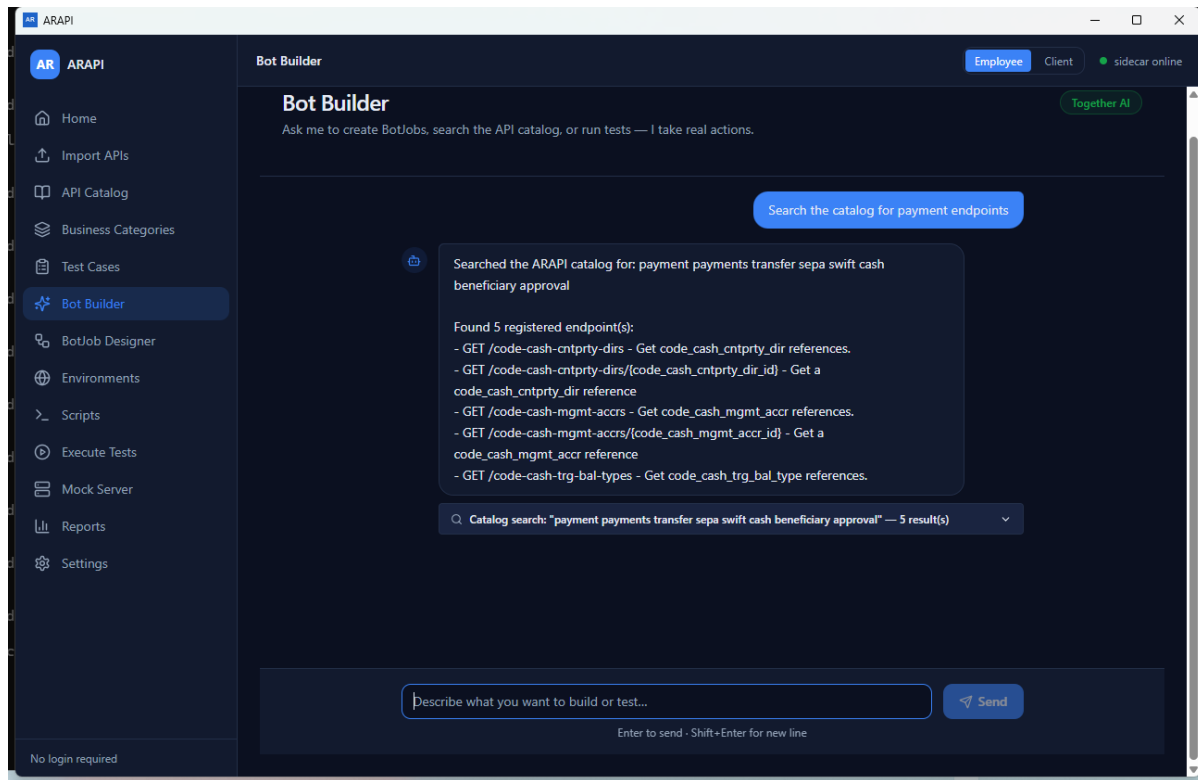


Figure 8. Bot Builder - catalog action

Review the returned endpoints and expand action cards. Never assume a conversational result is a production endpoint without catalog evidence.

Evidence rule: An action card is evidence of what ARAPI matched or created. Confirm whether the endpoint is imported or synthetic before relying on it.

6.9 BotJob Designer - no selection

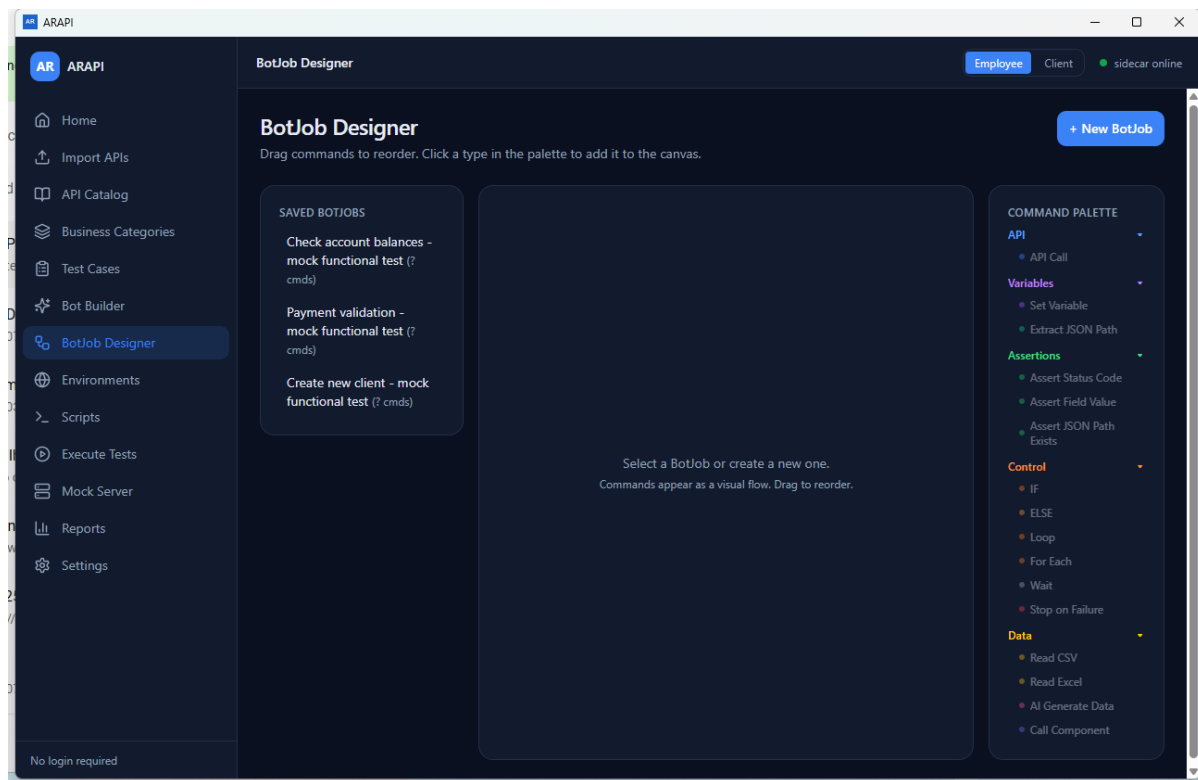


Figure 9. BotJob Designer - no selection

Select a saved BotJob or create one. Palette controls stay disabled until a job is selected.

6.10 BotJob Designer - selected job

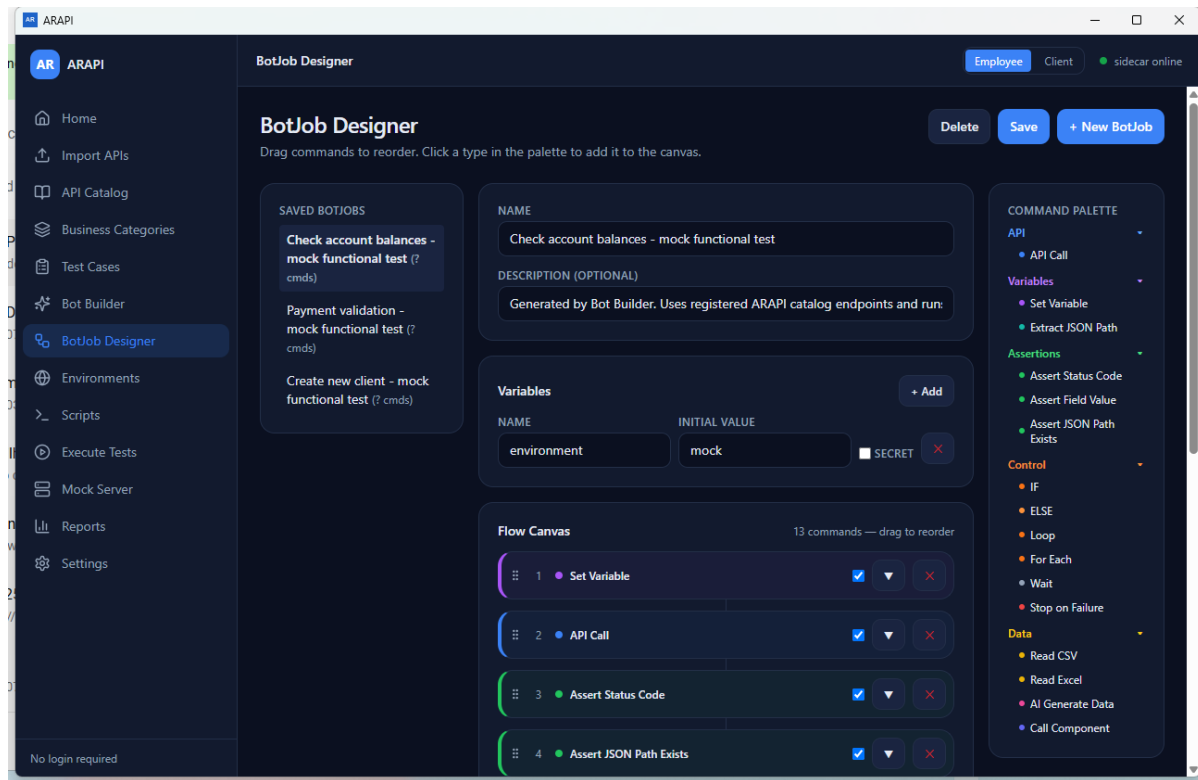


Figure 10. BotJob Designer - selected job

Edit metadata and variables, add palette commands, drag the grip to reorder, enable/disable commands, and Save.

6.11 BotJob Designer - API Call configuration

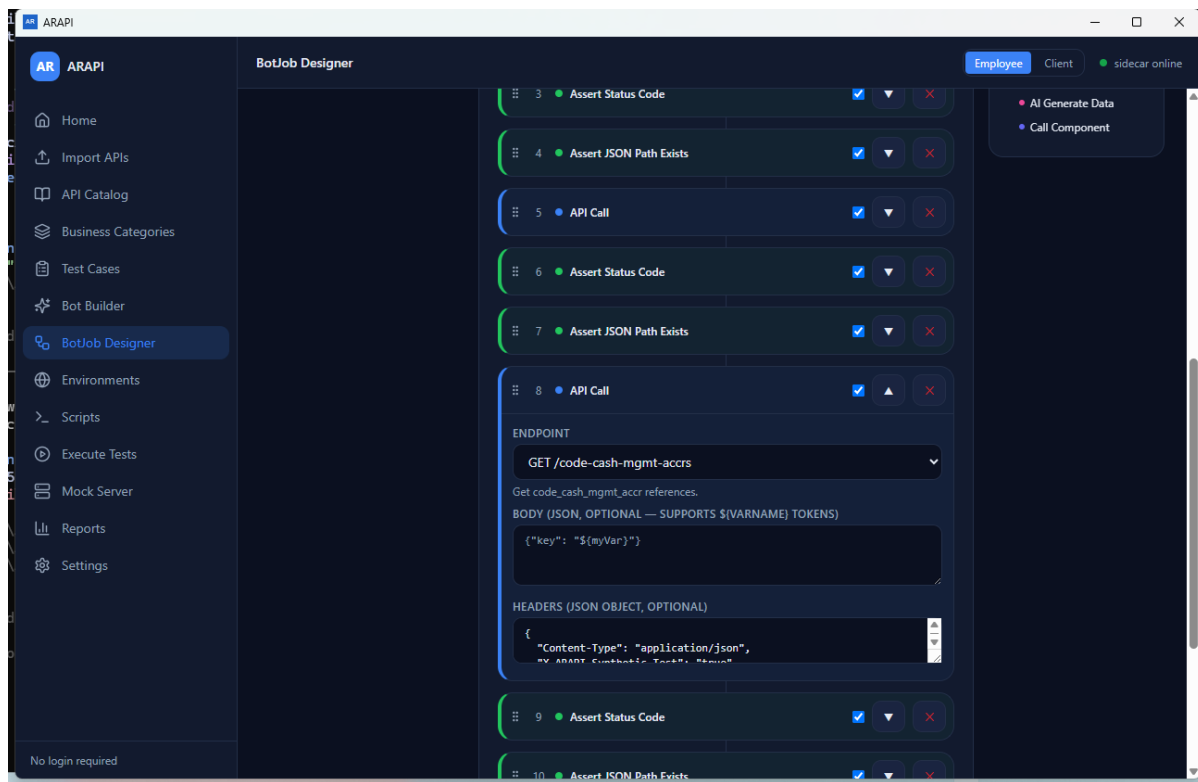


Figure 11. BotJob Designer - API Call configuration

Choose an imported endpoint, supply valid optional JSON, and reference variables with `${name}`.

```
${token}
{"Authorization": "Bearer ${token}"}
{"customerId": "${customerId}"}
```

6.12 Environments

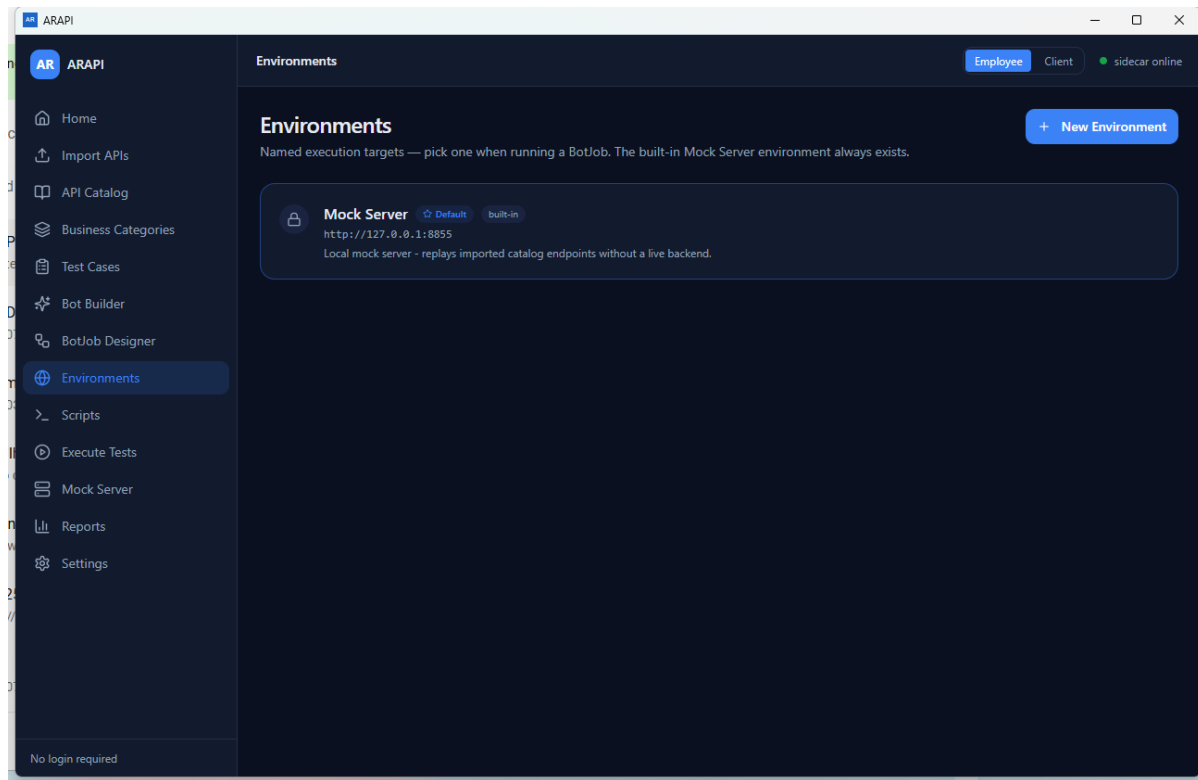
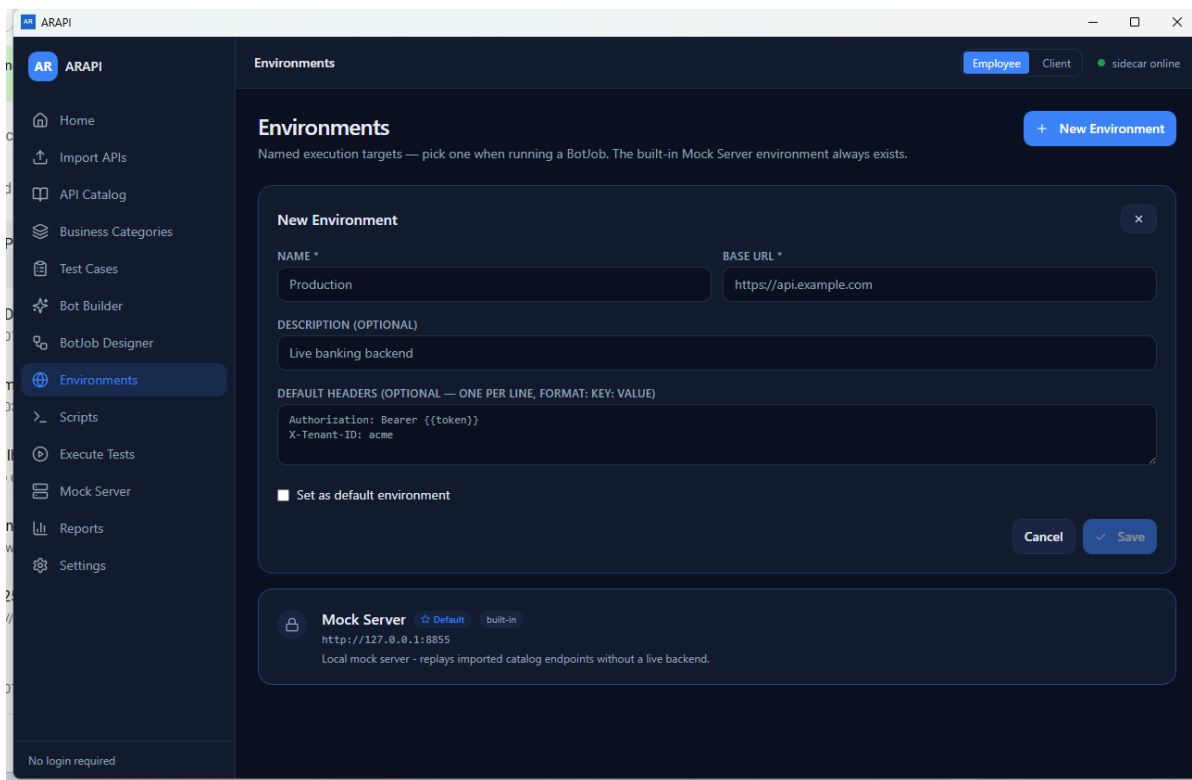


Figure 12. Environments

The built-in record cannot be edited or deleted. Create separate named targets for approved dev, test, or production systems.

6.13 Environment form



The screenshot shows the ARAPI application interface. On the left is a sidebar with navigation links: Home, Import APIs, API Catalog, Business Categories, Test Cases, Bot Builder, BotJob Designer, Environments (selected), Scripts, Execute Tests, Mock Server, Reports, and Settings. The main area is titled 'Environments' and includes a '+ New Environment' button. Below this is a 'New Environment' form with the following fields:

- NAME ***: Production
- BASE URL ***: https://api.example.com
- DESCRIPTION (OPTIONAL)**: Live banking backend
- DEFAULT HEADERS (OPTIONAL — ONE PER LINE, FORMAT: KEY: VALUE)**:
Authorization: Bearer {{token}}
X-Tenant-ID: acme
- ☒ **Set as default environment**

At the bottom of the form are 'Cancel' and 'Save' buttons. Below the form, a 'Mock Server' entry is shown with the URL 'http://127.0.0.1:8855' and the description 'Local mock server - replays imported catalog endpoints without a live backend.'

Figure 13. Environment form

Use one Key: Value header per line. Save is enabled only after Name and Base URL are present.

```
Authorization: Bearer ${token}
X-Tenant-ID: acme
```

6.14 Scripts - initial

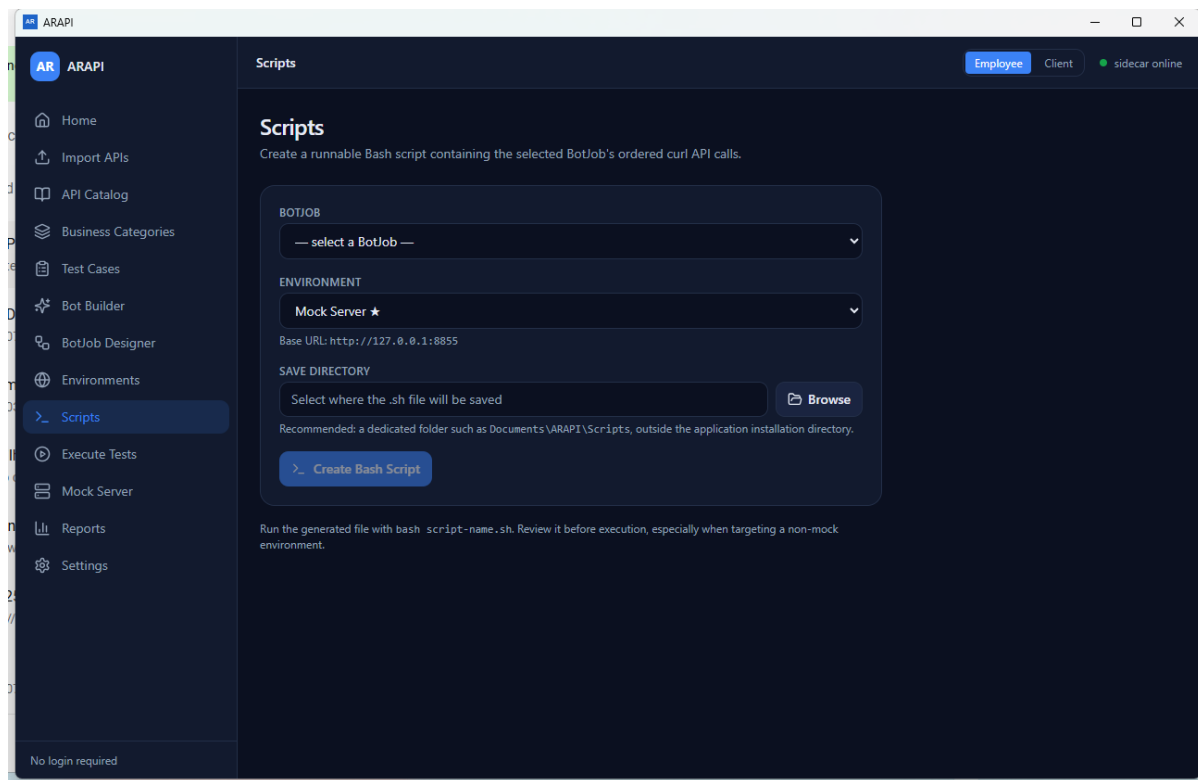


Figure 14. Scripts - initial

The action remains disabled until the required selections and desktop output directory are set.

6.15 Scripts - selected BotJob

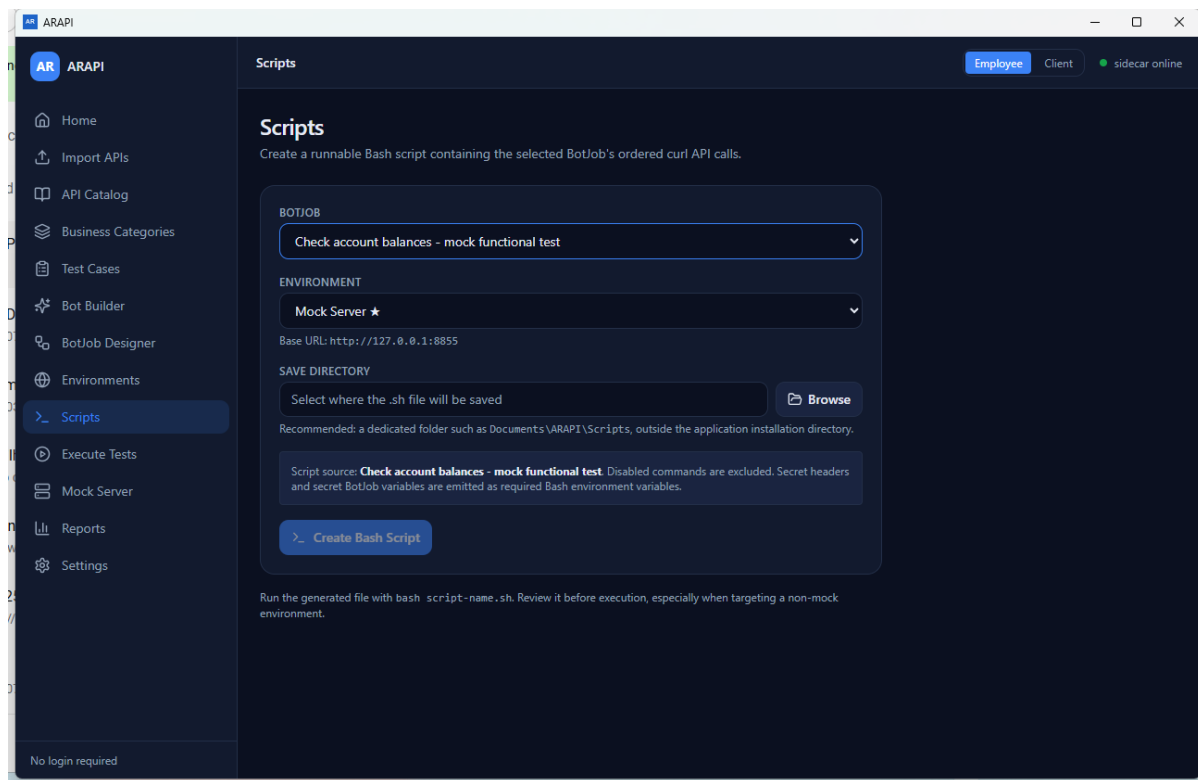


Figure 15. Scripts - selected BotJob

Choose Browse and use a dedicated directory such as Documents\ARAPI\Scripts. Then create the file.

Recommended directory: %USERPROFILE%\Documents\ARAPI\Scripts

6.16 Execute Tests - initial

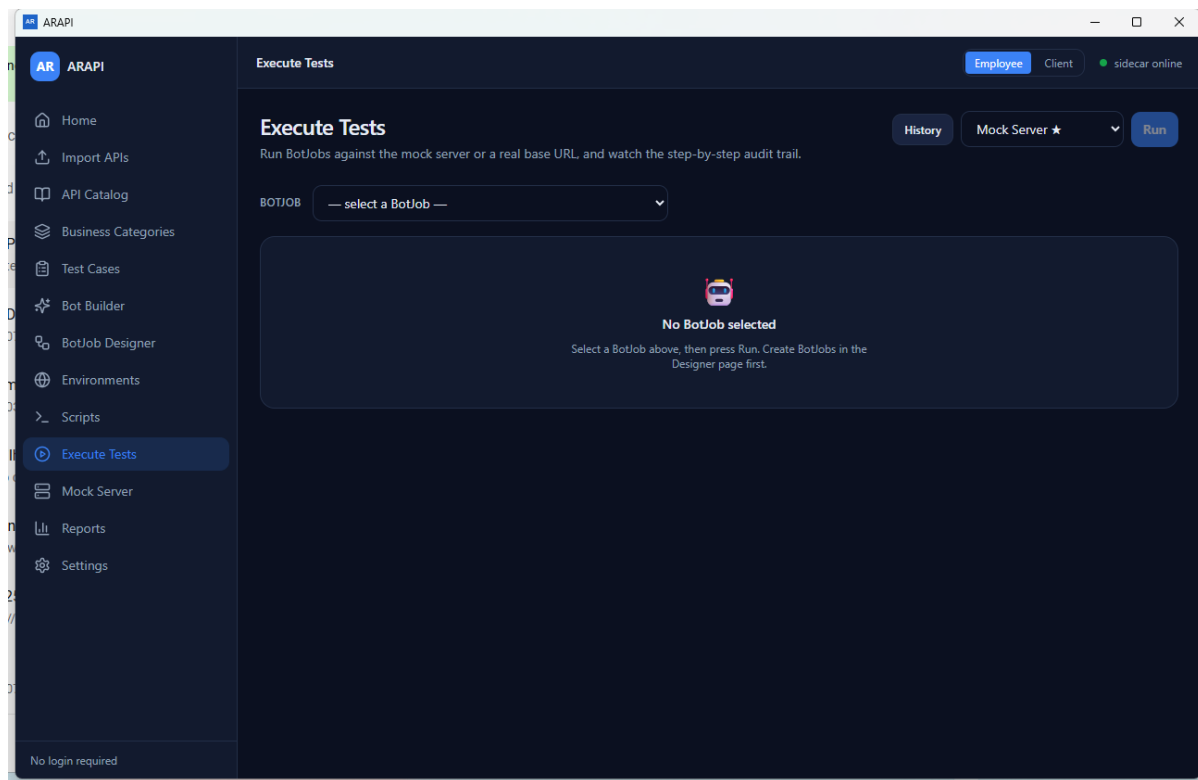


Figure 16. Execute Tests - initial

Select the execution environment and BotJob. Run and Create Scripts appear/enabled only when applicable.

6.17 Execute Tests - ready

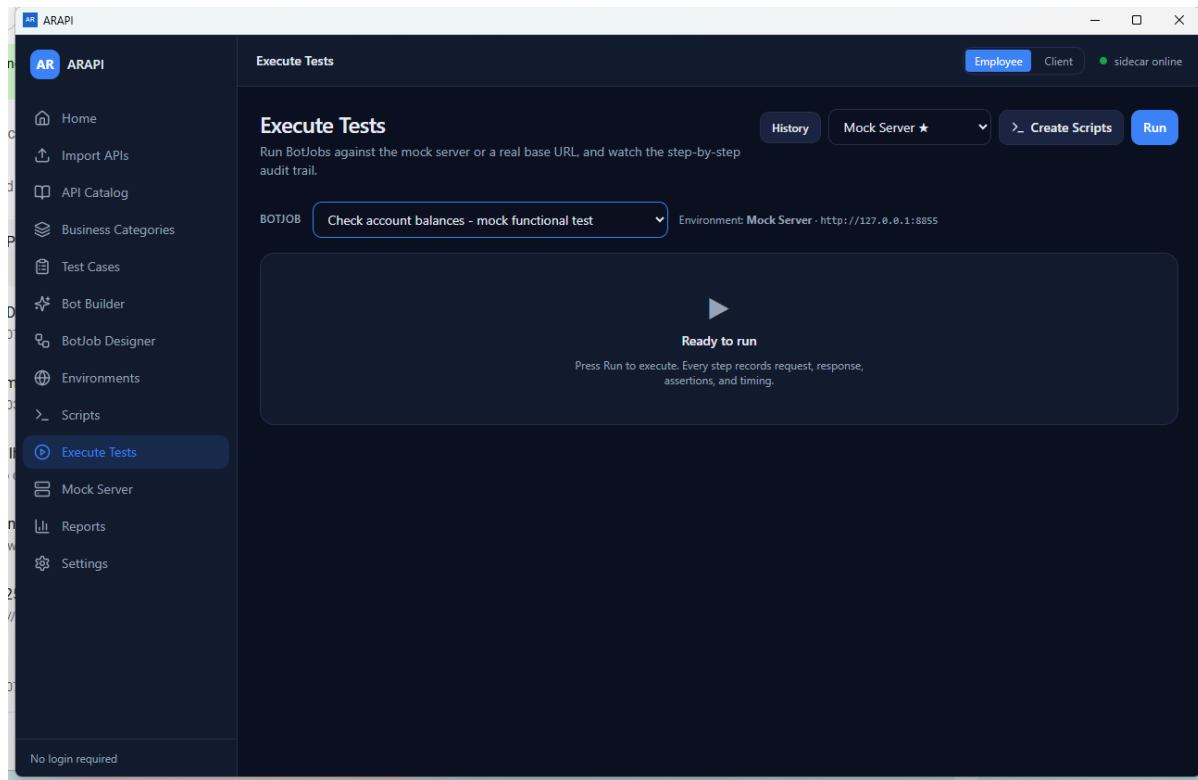


Figure 17. Execute Tests - ready

Create Scripts opens the Scripts page with the current BotJob and environment preselected; Run starts the local audit.

6.18 Execute Tests - result summary

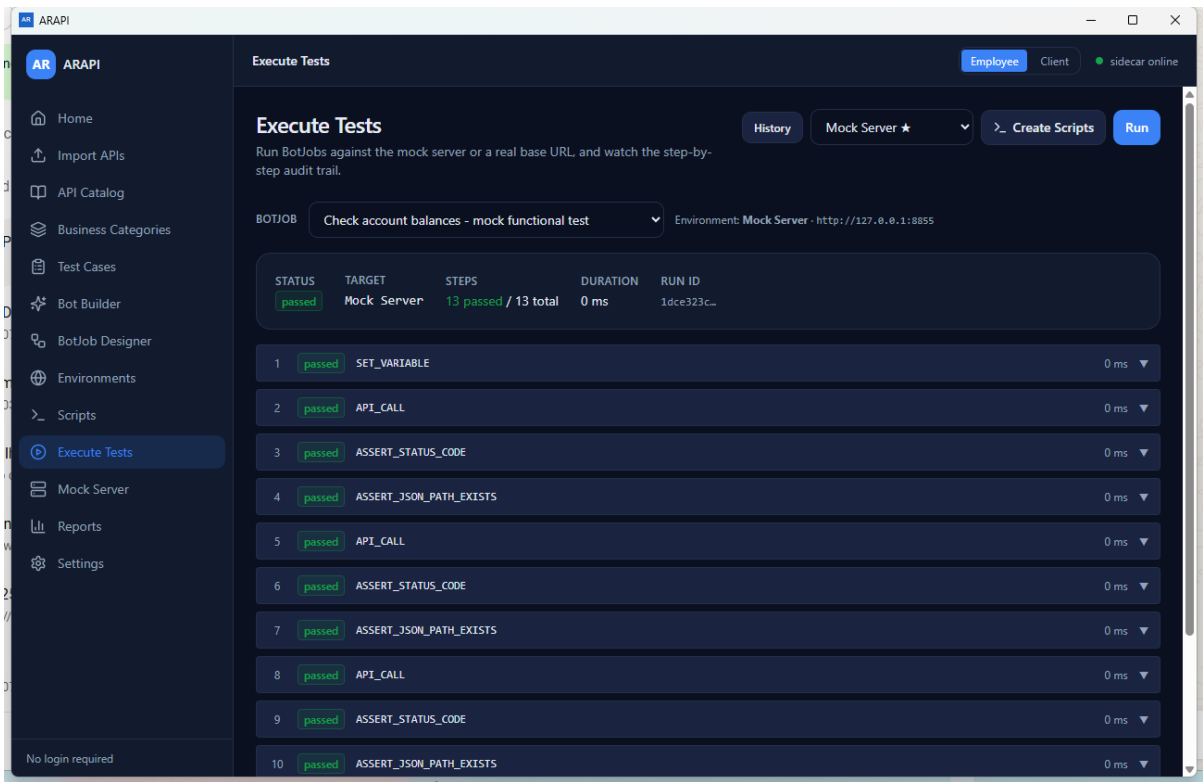


Figure 18. Execute Tests - result summary

Expand any row for details. Always correlate real-environment results with target-side evidence.

Evidence: Use target-side logs/data for live-system acceptance; a local ARAPI status alone is insufficient.

6.19 Execution step detail

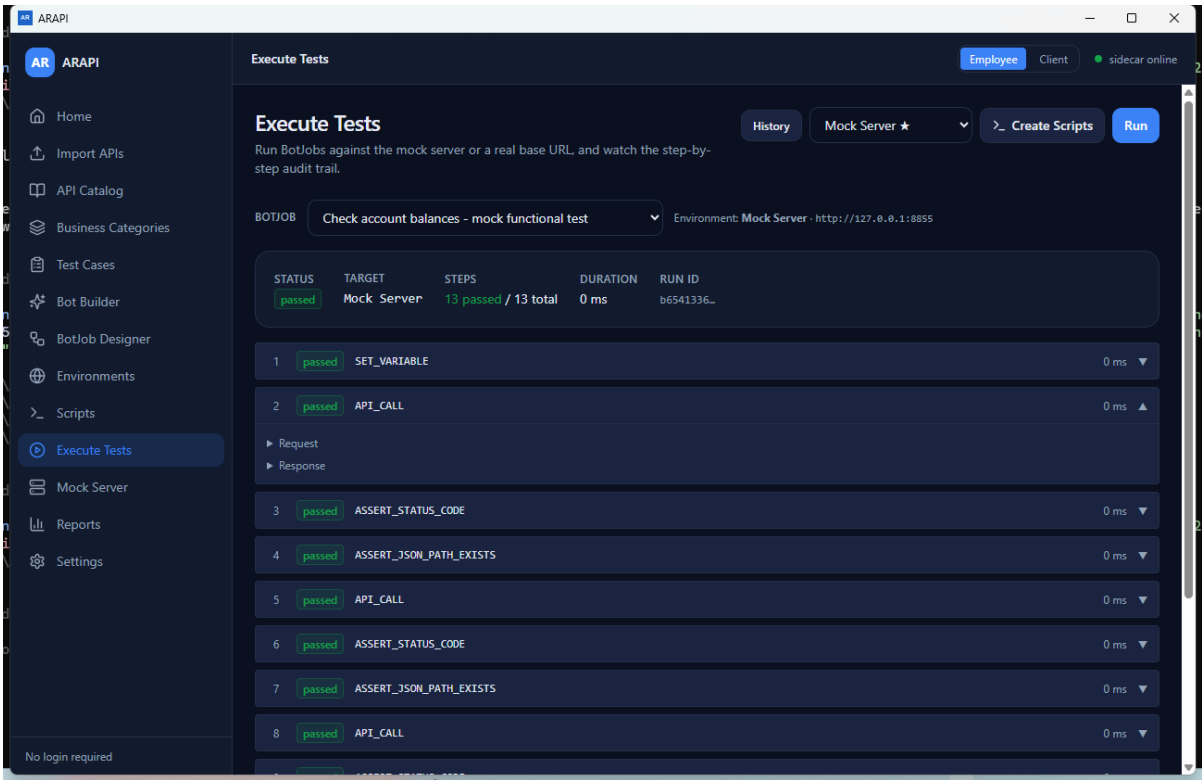


Figure 19. Execution step detail

Expand Request or Response when present. Do not copy sensitive payloads into tickets or client documents.

6.20 Execution History

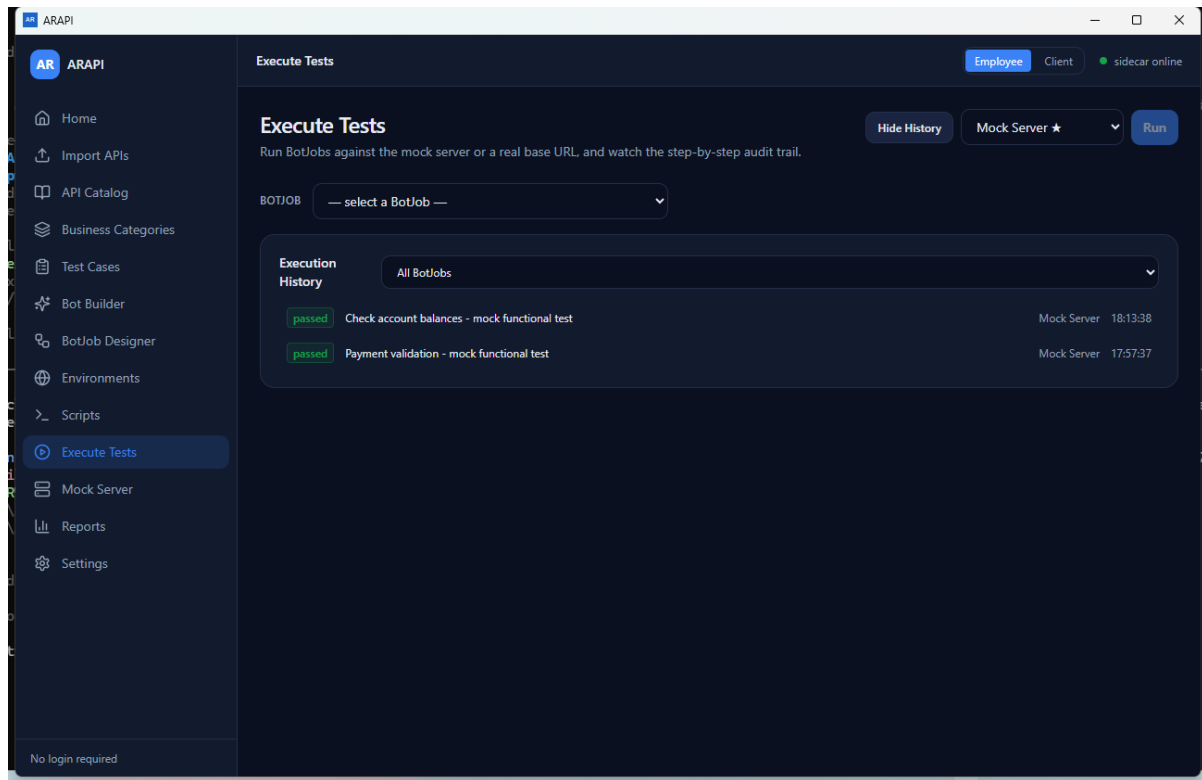


Figure 20. Execution History

Filter by BotJob and select a run to reload its summary and step results.

6.21 Mock Server - stopped

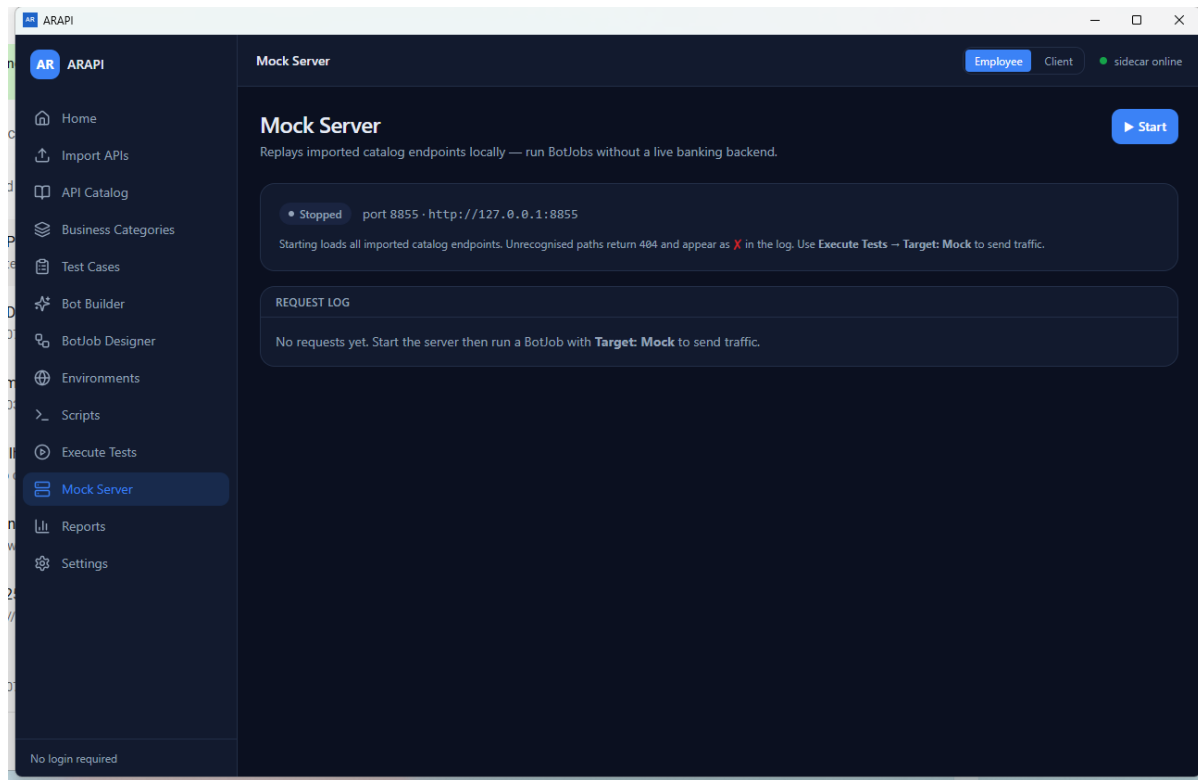


Figure 21. Mock Server - stopped

Start changes the ARAPI mock state and activates the dashboard log for the current build.

6.22 Mock Server - running

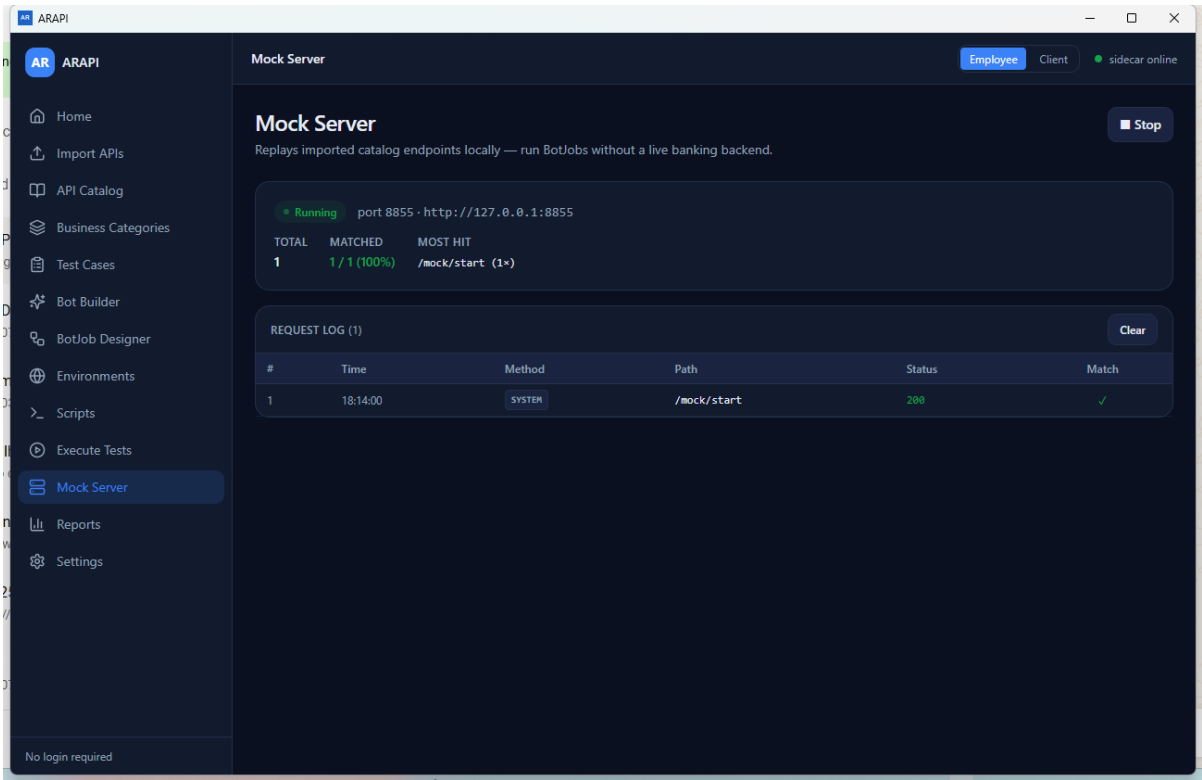


Figure 22. Mock Server - running

Stop changes state; Clear removes log rows. See the current-build note before treating port 8855 as an external listener.

Current build: The screenshot proves ARAPI state/log changes. It does not by itself prove that an external listener is bound to port 8855.

6.23 Reports and Exports

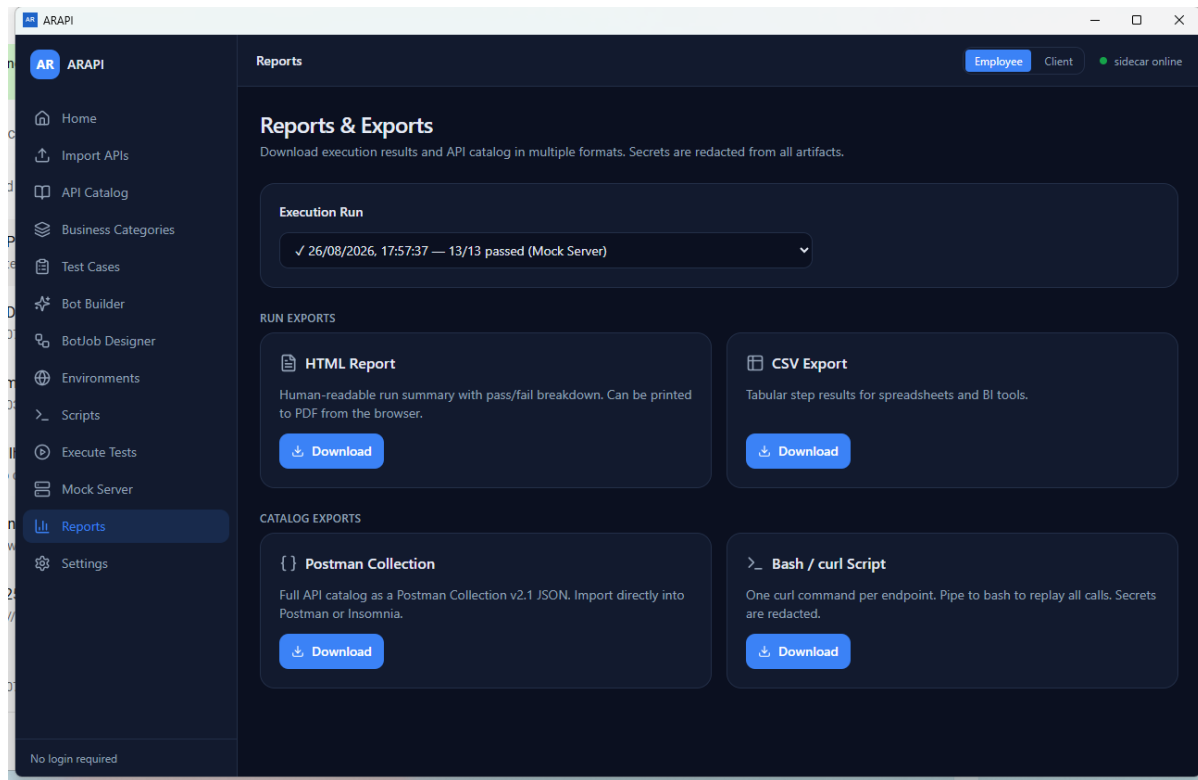


Figure 23. Reports and Exports

Run exports depend on a selected execution. Catalog exports are independent and cover the whole catalog.

Export distinction: Run HTML/CSV uses one selected run. Postman and Bash/curl export the whole catalog and default to a localhost base URL unless an integration caller supplies one.

6.24 Settings - provider list

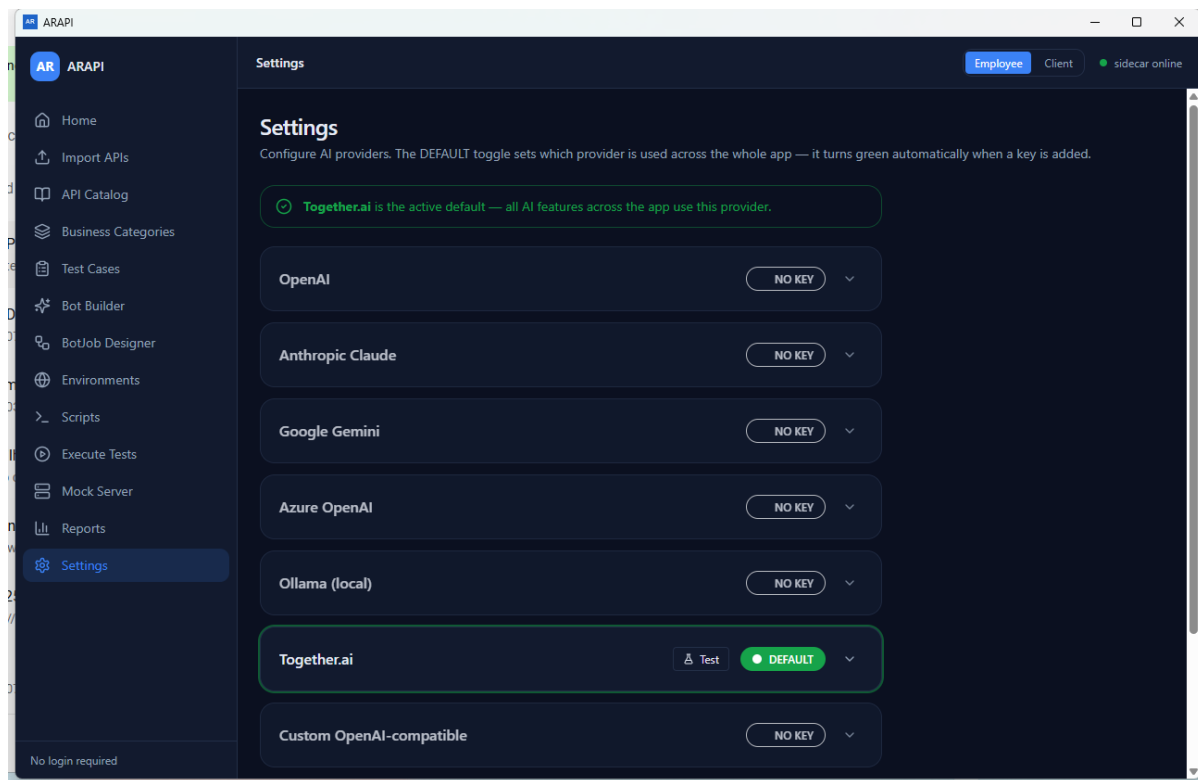


Figure 24. Settings - provider list

A green **DEFAULT** badge identifies the provider used by AI features. **NO KEY** providers cannot be selected as default.

6.25 Settings - provider form

ARAPI

Settings

Employee Client sidcar online

Settings

Configure AI providers. The DEFAULT toggle sets which provider is used across the whole app — it turns green automatically when a key is added.

✔ Together.ai is the active default — all AI features across the app use this provider.

OpenAI NO KEY ^

MODEL (OPTIONAL)

gpt-4o-mini

API KEY

Enter API key

Encrypted at rest (AES-256-GCM) — never returned to the browser.

Save

Anthropic Claude NO KEY v

Google Gemini NO KEY v

Azure OpenAI NO KEY v

No login required

Figure 25. Settings - provider form

Enter the model and required provider values, save, then use Test as a configuration indicator.

Credential handling: The key field is masked and provider-list responses omit key values. Still protect the local ARAPI data store as sensitive and use non-production credentials where possible.

6.26 Settings - local services

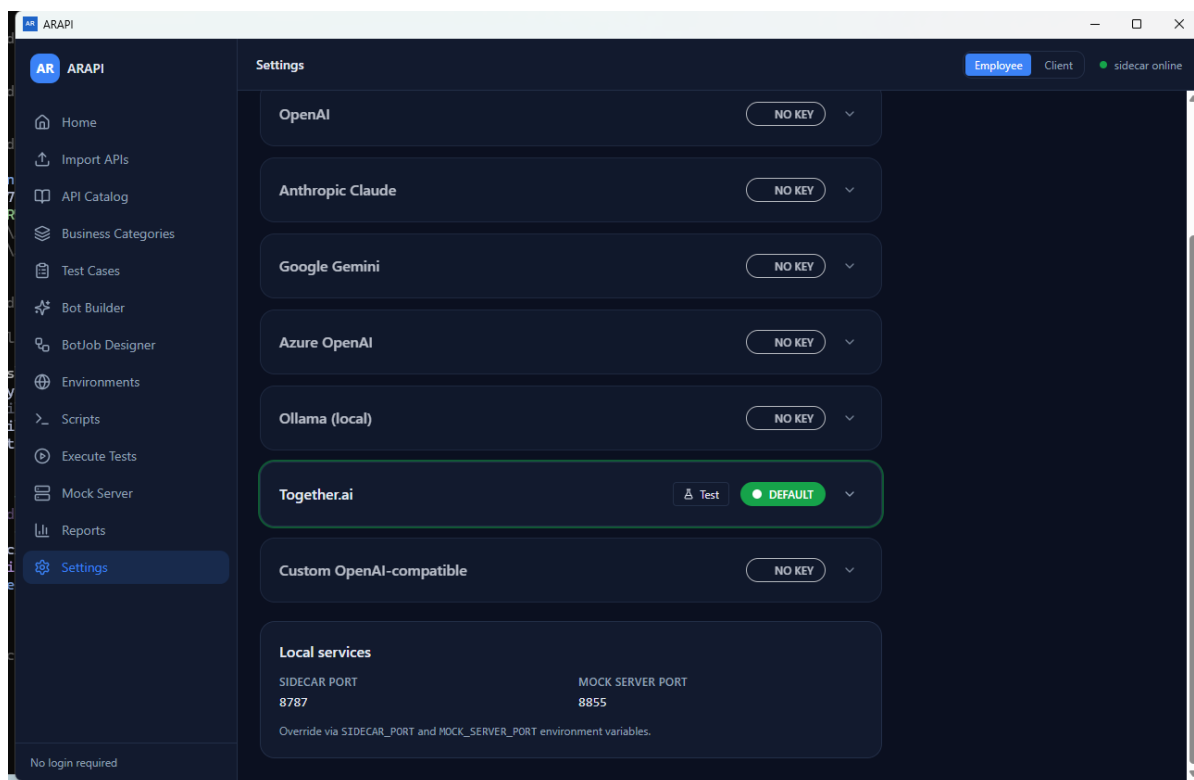


Figure 26. Settings - local services

These ports are localhost services. Environment-variable overrides are administrator/deployment settings.

7. BotJob command reference

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Commands execute in canvas order when enabled. Click a palette item to add it, use the grip to reorder, the checkbox to enable/disable, the chevron to edit, and x to delete. Save after every material change.

Bash boundary: Only enabled API_CALL commands are exported to the BotJob-specific .sh file. The other command descriptions below apply to the Designer/runtime model and local run audit.

7.1 API commands

API Call (API_CALL)

Availability: Configurable

Purpose: Select one real endpoint from the imported catalog and prepare an HTTP request.

Configuration: Endpoint; optional JSON body; optional JSON header object. Body and header values accept \${variable} tokens.

Current release: Included by BotJob Bash export when enabled. In the run audit, request/response details are available when the runtime records them.

7.2 Variables commands

Set Variable (SET_VARIABLE)

Availability: Configurable

Purpose: Create or replace a runtime value for later commands.

Configuration: Variable name and value. The value may contain \${variable} tokens.

Current release: Not written as a separate Bash statement by the BotJob exporter; declared BotJob variables drive Bash defaults and guards.

Extract JSON Path (EXTRACT_JSON_PATH)

Availability: Configurable

Purpose: Read a value from the most recent JSON response and store it in a variable.

Configuration: JSON path such as \$.data.token and destination variable name.

Current release: Available in the Designer/runtime model; not converted to Bash by the current exporter.

7.3 Assertions commands

Assert Status Code (ASSERT_STATUS_CODE)

Availability: Configurable

Purpose: Compare the last response status with an expected HTTP code.

Configuration: Expected integer from 100 through 599, commonly 200 or 201.

Current release: Appears in the local run audit; not converted to Bash by the current exporter.

Assert Field Value (ASSERT_FIELD_VALUE)

Availability: Configurable

Purpose: Compare a JSON-path value in the last response with an expected JSON value.

Configuration: JSON path and expected value. Valid JSON input is parsed; otherwise it is treated as text.

Current release: Appears in the local run audit; not converted to Bash by the current exporter.

Assert JSON Path Exists (ASSERT_JSON_PATH_EXISTS)

Availability: Configurable

Purpose: Confirm that the requested JSON path exists in the last response.

Configuration: JSON path such as \$.data.id.

Current release: Appears in the local run audit; not converted to Bash by the current exporter.

7.4 Control commands

Wait (WAIT)

Availability: Configurable

Purpose: Pause the workflow for a bounded delay.

Configuration: Milliseconds from 0 through 10,000.

Current release: Recognized by the execution-engine model; not converted to Bash by the current exporter.

Stop on Failure (STOP_ON_FAILURE)

Availability: Verify in deployment

Purpose: Mark the workflow to stop after a failure.

Configuration: No visible fields.

Current release: The Bash exporter continues remaining curl calls, counts failures, and exits 1 at the end. Acceptance-test desktop stop behavior before relying on it.

IF (IF)

Availability: Preview

Purpose: Reserved conditional branch start.

Configuration: No configuration editor in this release.

Current release: Designer warns that this command is not executed and is recorded as skipped.

ELSE (ELSE)

Availability: Preview

Purpose: Reserved alternate branch.

Configuration: No configuration editor in this release.

Current release: Not executed; recorded as skipped.

Loop (LOOP)

Availability: Preview

Purpose: Reserved repeated execution block.

Configuration: No configuration editor in this release.

Current release: Not executed; recorded as skipped.

For Each (FOR_EACH)

Availability: Preview

Purpose: Reserved iteration over a collection.

Configuration: No configuration editor in this release.

Current release: Not executed; recorded as skipped.

7.5 Data commands

Read CSV (READ_CSV)

Availability: Preview

Purpose: Reserved CSV-driven data input.

Configuration: No configuration editor in this release.

Current release: Not executed; recorded as skipped.

Read Excel (READ_EXCEL)

Availability: Preview

Purpose: Reserved spreadsheet-driven data input.

Configuration: No configuration editor in this release.

Current release: Not executed; recorded as skipped.

AI Generate Data (AI_GENERATE_DATA)

Availability: Preview

Purpose: Reserved AI-assisted test-data creation.

Configuration: No configuration editor in this release.

Current release: Not executed; recorded as skipped.

Call Component (CALL_COMPONENT)

Availability: Preview

Purpose: Reserved call to a reusable component/test flow.

Configuration: No configuration editor in this release.

Current release: Not executed; recorded as skipped. The Test Cases page is also reserved in this release.

Command design rules

- Place API_CALL before assertions or extraction that depend on its response.
- Use catalog endpoint selection; do not paste an unregistered path into descriptive fields.
- Use \${name} consistently and declare variables with meaningful names.
- Keep secret initial values empty. Mark the variable secret so Bash requires an environment value.
- Disable rather than delete a command when temporarily excluding it for diagnosis.
- Do not rely on preview commands in client acceptance criteria.

8. Variables and environments

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Variables

A BotJob variable has a name, optional initial value, and secret flag. Reference it in supported command text with \${name}.

Element	Guidance
Name	Use letters, digits, and underscores; prefer clear business names such as customerId or accessToken.

Element	Guidance
Initial value	Use only non-sensitive defaults. Leave secret values empty.
secret	For Bash export, requires the normalized shell variable from the environment.
Normalization	Bash names become uppercase; non-alphanumeric characters become underscores; a leading digit receives ARAPI_.
Unknown token	The exporter treats the normalized token as required and stops before curl if it is unset.

```
customer-id -> CUSTOMER_ID
2fa-token   -> ARAPI_2FA_TOKEN
${accountId} -> ${ACCOUNTID} (normalized by the generated script)
```

Environments

An environment combines a base URL, optional description, and default headers. One environment can be marked default; the built-in Mock Server environment always exists and cannot be deleted.

- Name and Base URL are mandatory.
- Enter default headers one per line as Key: Value.
- Command headers override environment headers with the same name in Bash generation.
- Sensitive header names without a \${token} are replaced by required Bash environment variables rather than embedded in the generated file.
- Environment cards display saved headers in the current UI. Do not save literal production secrets as environment headers.

Safe pattern: Store Authorization: Bearer \${token}, declare token as a secret variable with no initial value, and supply TOKEN only to the Bash process or approved CI secret store.

9. Bash/curl script generation and execution

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Generate from Scripts

1. Select the saved BotJob.
2. Select the environment and verify the displayed base URL.
3. Select Browse and choose a dedicated folder such as Documents\ARAPI\Scripts.

4. Select Create Bash Script. ARAPI writes a timestamped file such as check-account-balances-20260826-181500.sh.
5. Read the complete file before running it, especially for POST, PUT, PATCH, or DELETE calls.

Generate from Execute Tests

After selecting a BotJob, the conditional Create Scripts button opens Scripts with the current BotJob and environment preselected. You still choose the save directory and select Create Bash Script.

What the exporter includes

- Enabled API_CALL commands in their saved order.
- The selected environment base URL and merged environment/command headers.
- JSON request bodies and \${variable} interpolation.
- Path parameters such as {accountId} converted to required shell variables.
- Required-variable guards for secret variables, sensitive headers, path parameters, and unknown tokens.
- A failure counter: each failed curl is reported, remaining calls continue, and the script exits 1 if any call failed.

Not included: Assertions, extraction, Set Variable steps, waits, stop markers, preview control flow, data-file commands, and component calls are not emitted into the BotJob-specific Bash file.

Generated Bash statement reference

Statement	Meaning
#!/usr/bin/env bash	Runs the file with Bash found in the environment.
# Generated by ARAPI...	Documents the source BotJob/environment; comments do not execute.
set -uo pipefail	Treats unset variables as errors and propagates pipeline failures. The exporter intentionally handles curl failures itself.
DEFAULT_BASE_URL='...'	Stores the selected environment URL as the default.
BASE_URL="\${BASE_URL:-\$DEFAULT_BASE_URL}"	Allows a runtime BASE_URL override without editing the file.
failures=0	Initializes the curl failure counter.
: "\${TOKEN:?Set TOKEN...}"	Stops immediately when a required value is unset or empty.
DEFAULT_NAME='value'	Stores a non-secret BotJob initial value.
NAME="\${NAME:-\$DEFAULT_NAME}"	Allows an environment override or uses the non-secret default.

Statement	Meaning
echo '==> n/N METHOD /path'	Prints progress before each call.
if ! curl \	Runs curl and enters the failure branch when curl returns non-zero.
--globoff	Prevents curl from treating braces/brackets in URLs as glob patterns.
--fail-with-body	Returns failure for HTTP 4xx/5xx while retaining the response body.
--silent --show-error	Suppresses progress meter but keeps error messages.
--request METHOD	Uses the catalog HTTP method.
--header 'Key: value'	Adds merged environment and command headers.
--data 'JSON'	Sends the optional body and causes application/json to be added when Content-Type was absent.
"\${BASE_URL}"/path'	Combines runtime/default base URL with the catalog path.
failures=\$((failures + 1))	Counts a failed curl while allowing the next API call.
if ((failures > 0)); then ... exit 1	Returns a non-zero process result when any API call failed.
echo 'All API calls completed successfully.'	Prints only when the failure counter is zero.

Run on Windows with Git Bash

```
cd /c/Users/YOUR_NAME/Documents/ARAPI/Scripts
export TOKEN='temporary-secret-value'
export ACCOUNT_ID='12345'
bash ./script-name.sh
echo $?
```

Run from Windows PowerShell when bash is on PATH

```
$env:TOKEN = 'temporary-secret-value'
$env:ACCOUNT_ID = '12345'
$env:BASE_URL = 'https://approved-test-api.example'
bash ./script-name.sh
$LASTEXITCODE
Remove-Item Env:TOKEN, Env:ACCOUNT_ID
```


Run from Windows Command Prompt

```
cd /d "%USERPROFILE%\Documents\ARAPI\Scripts"
set "TOKEN=temporary-secret-value"
set "ACCOUNT_ID=12345"
bash script-name.sh
echo %ERRORLEVEL%
set "TOKEN="
set "ACCOUNT_ID="
```

Run with Windows Subsystem for Linux

```
cd /mnt/c/Users/YOUR_NAME/Documents/ARAPI/Scripts
export TOKEN='temporary-secret-value'
bash ./script-name.sh
echo $?
```

Pre-run review checklist

- Correct BotJob name and timestamped file.
- Correct BASE_URL and approved environment.
- No unintended destructive methods or paths.
- Headers/bodies contain no embedded secret or real customer data.
- All required shell variables are set from an approved source.
- curl is available, TLS trust is configured, and network routing is approved.
- Expected side effects, monitoring, and rollback are agreed before execution.

10. Execution results, history, and reports

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Run audit

Execute Tests stores a run summary and step rows. Summary fields are Status, Target, Steps, Duration, and Run ID. Step rows show command type, status, duration, and expandable details when recorded.

Status	Interpretation
passed	The local run record marks the step/run successful.
failed	An assertion or step reported a failure.
error	The step could not complete normally, for example a missing endpoint reference.
skipped	The command was disabled, stopped, preview-only, or otherwise not executed.

Status	Interpretation
running	The run has not yet reached its final state.

Acceptance rule: For a real environment, require target-side logs, database/state evidence, or an approved response capture in addition to the ARAPI run status.

History

1. Select History to show saved runs.
2. Use All BotJobs to filter by workflow.
3. Select a row to reload the summary and steps; the panel closes automatically.

Reports

Export	Scope	Use
HTML Report	Selected run	Readable summary; can be printed to PDF by the user's browser/viewer.
CSV Export	Selected run	Tabular step results for spreadsheet or BI review.
Postman Collection	Whole catalog	Postman Collection v2.1 JSON; review the default localhost base URL.
Bash / curl Script	Whole catalog	One simple curl per endpoint; unlike Scripts, it is not BotJob-specific.

Prefer Scripts for controlled execution: The Scripts page preserves a selected BotJob's enabled API_CALL order, environment headers, body data, variable guards, and final failure status. Reports Bash/curl is a broad catalog export.

11. Bot Builder and AI providers

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Bot Builder

Bot Builder sends the visible conversation history to the local backend, which can search the catalog, list/create BotJobs, and request mock runs. Action cards expose catalog matches, created jobs, and run summaries.

- Use a suggestion or enter a precise business flow.
- Review catalog-search evidence before accepting a proposed workflow.

- If the result says a synthetic mock endpoint/data record was created, keep it out of production acceptance and exports intended to represent real contracts.
- Open Designer and review the created job, endpoint selections, body, headers, variables, and command order before Save or execution.
- Do not paste credentials, customer identifiers, account data, or confidential payloads into the conversation.

Supported provider records

Provider	Default model shown by ARAPI	Base URL field
OpenAI	gpt-4o-mini	Not shown
Anthropic Claude	claude-3-5-haiku-20241022	Not shown
Google Gemini	gemini-1.5-flash	Not shown
Azure OpenAI	gpt-4o-mini	Required by deployment
Ollama (local)	llama3.2	Shown; key not required for local Ollama
Together.ai	meta-llama/Llama-3-8b-chat-hf	Not shown
Custom OpenAI-compatible	gpt-4o-mini	Required by deployment

Configure a provider

1. Expand the provider card.
2. Enter or confirm the model and base URL where shown.
3. Enter the API key and select Save. The field is masked; a blank key on later saves preserves the existing key.
4. Select SET DEFAULT on a configured provider. If exactly one key exists, ARAPI may promote it automatically.
5. Select Test, then make an actual Bot Builder request to verify end-to-end connectivity.

Local data sensitivity: Provider-list responses omit key values, but the workstation and ARAPI data directory must still be protected as credential-bearing assets. Use scoped, revocable, non-production keys for testing.

12. Security and operational practices

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Before use

- Use an organization-managed Windows account and approved ARAPI distribution.
- Keep the local backend bound to localhost; do not expose service ports publicly.
- Back up the correct ARAPI data directory before upgrades or data-cleanup work.
- Import specifications only from trusted sources and review import failures.
- Create separate mock/dev/UAT/production environments and label them clearly.

Credentials and data

- Never store literal secrets in BotJob initial values, non-secret headers, descriptions, screenshots, or client documents.
- Use secret BotJob variables and short-lived environment variables or an approved CI secret store.
- Generated scripts and exported reports may contain paths, bodies, headers, response details, and business metadata; classify and retain them accordingly.
- Review scripts after generation and again before every non-mock run.
- Clear shell environment variables and delete temporary artifacts according to policy after use.

Change control

- Record ARAPI version, BotJob name, environment, script filename/hash, operator, start/end time, and target approval.
- Use a least-privilege API identity with a narrow expiration window.
- For write operations, document expected records and rollback/compensation before execution.
- Preserve HTML/CSV and target-side evidence in the approved audit location; do not rely on screenshots alone.

No-login does not mean no security: ARAPI's desktop UI has no internal login. Operating-system access, file ACLs, secret management, API authorization, and audit retention provide the security boundary.

13. Troubleshooting

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Use the exact displayed error, affected BotJob/environment, ARAPI version, and reproduction steps when escalating. Remove secrets and customer data from attachments.

sidecar offline

Close and reopen ARAPI. Confirm no other process is occupying port 8787. If the status stays offline, contact the deployment administrator; the UI cannot work without the local backend.

Import is disabled

Choose at least one supported file on Upload Files, or select a valid directory on Folder (Desktop). Supported extensions are .yaml, .yml, and .json.

Import reports zero endpoints

Confirm the files contain OpenAPI/Swagger path operations, not only schemas/examples. Review the failure list and try one known-good specification.

Repeated catalog rows

Avoid importing the same specification tree repeatedly when duplicate rows are observed. Record the source folder and contact support before cleaning shared client data.

Palette commands are dimmed

Select or create a BotJob first.

Save does not enable

For environments, Name and Base URL are mandatory. Header lines must use Key: Value. For scripts, job, environment, and desktop output directory are mandatory.

API call references a missing endpoint

Open Designer, expand the API Call, select the current endpoint from the catalog, and Save. This can occur after catalog content is replaced.

Create Bash Script says there are no calls

Enable at least one API_CALL and ensure it references an endpoint still present in the catalog.

bash is not recognized

Install/use Git Bash, Windows Subsystem for Linux, or a CI agent with Bash and curl. Windows cmd.exe alone cannot interpret Bash syntax.

Script stops before the first curl

A required variable guard failed. Set the named environment variable, including path parameters and sensitive headers, then run again.

Script ends with exit code 1

One or more curl calls failed. Scroll to the first curl error, verify BASE_URL, network/TLS/authentication, and review the response body.

Run results are green but target evidence is absent

Treat ARAPI's local run summary as an audit record, not independent proof of a downstream state change. Use the generated Bash script and target-side logs/data for acceptance.

Port 8855 cannot be reached

The observed 1.0.0 C# desktop build maintains mock state/log in ARAPI. Do not assume an external listener is available unless your deployment administrator has enabled and verified one.

Reports has no run selector

Execute a BotJob first. Catalog Postman/Bash exports remain available without a run.

AI provider Test shows success but AI calls fail

The current desktop Test is a local configuration indicator. Verify endpoint, model, key entitlement, proxy/firewall, and an actual Bot Builder request.

Diagnostic information safe to collect

- ARAPI version and whether installer or portable distribution is used.
- Home engine status and Settings local-service port values.
- Specification filename/type and import counts (not confidential contents unless approved).
- BotJob name, command types/order, and whether the failing command is enabled.
- Environment name and sanitized base URL; never send tokens or full Authorization headers.
- Script filename, exit code, and sanitized curl error text.
- Run ID and report with sensitive request/response content removed.

14. Feature status matrix

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Capability	Status	Qualification
Import YAML/JSON files/folders	Available	Recursive upload/desktop folder workflows.
Catalog search and mapping display	Available	Mapping is shown; no mapping editor button on the Catalog page.
25-domain banking taxonomy	Available	Read-only cards in current UI.
Test Case creation	Reserved	New test case is disabled.
Bot Builder catalog/search/create/run actions	Available with qualifications	May use synthetic mock endpoint/data when no strong catalog match.
Designer core commands	Configurable	API, variables, assertions, extract, wait; verify runtime acceptance per deployment.
Designer control/data preview commands	Preview	IF/ELSE/loops/read files/AI data/component are recorded skipped.
Environment CRUD	Available	Built-in Mock cannot be deleted.
BotJob-specific Bash/curl	Available	Enabled API_CALL commands only; desktop exact-directory picker.
Execute Tests audit/history	Available with qualification	Local run evidence; correlate real-system effects externally.
Mock dashboard	State/log available	External port listener must be deployment-verified.
HTML/CSV run reports	Available	Requires a saved run.
Postman/whole-catalog Bash	Available	Defaults to localhost base URL via UI download.
AI provider records/default selection	Available	Protect local data store as sensitive.
AI provider Test	Configuration indicator	Use a real Bot Builder request for end-to-end proof.

Appendix A. Complete control index

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This index covers every visible button, selector, disclosure control, status control, and important conditional action in the ARAPI 1.0.0 UI.

Page	Control	Available when	Result
Global	Employee / Client	Always	Changes the shared conversation-mode context; it does not rewrite catalog or BotJobs.
Global	sidecar online/offline	Always	Displays local backend health. Resolve offline status before doing work.
Home	Shortcut cards	Always	Open the matching sidebar destination.
Import APIs	Upload Files tab	Always	Shows file/folder upload workflow.
Import APIs	Folder (Desktop) tab	Always	Shows local folder-path workflow.
Import APIs	Drop area	Upload tab	Adds dropped .yaml, .yml, and .json files; subfolders are scanned.
Import APIs	Browse Files	Upload tab	Opens a multi-file picker.
Import APIs	Browse Folder	Upload tab	Opens a folder picker and adds supported files below it.
Import APIs	Remove (x)	After adding a file	Removes one staged item before import.
Import APIs	Import	After files/path selected	Imports specifications and reports specification, endpoint, and failure counts.
Import APIs	Browse	Desktop folder tab	Selects one local folder.
API Catalog	Search method / path	Always	Filters by method, path, or summary.
Test Cases	New test case	Disabled in 1.0.0	Reserved; does not create a record in this release.
Bot Builder	Suggestion buttons	Empty conversation	Sends the displayed prompt immediately.
Bot Builder	Catalog action card	After search action	Expands/collapses endpoint evidence.
Bot Builder	Open Designer	After BotJob creation action	Navigates to Designer; select the created job if it is not automatically highlighted.
Bot Builder	View Results	After execution action	Navigates to Execute Tests.
Bot Builder	Send	When prompt is non-empty	Sends the conversation; Enter also sends and Shift+Enter adds a line.
BotJob Designer	+ New BotJob	Always	Prompts for a name and creates a saved BotJob shell.
BotJob Designer	Saved BotJob row	When jobs exist	Loads the selected job and commands.
BotJob Designer	Delete	Job selected	Deletes the BotJob after confirmation.
BotJob Designer	Save	Job selected	Persists metadata, variables, order, configuration, and enabled state.
BotJob Designer	+ Add variable	Job selected	Adds a variable row.

Page	Control	Available when	Result
BotJob Designer	secret checkbox	Variable exists	Causes Bash generation to require the value from the environment rather than embed an initial value.
BotJob Designer	Variable delete (x)	Variable exists	Removes the variable from the working definition.
BotJob Designer	Palette group heading	Always	Expands or collapses one command group.
BotJob Designer	Palette command	Job selected	Adds the command to the end of the canvas and opens its editor.
BotJob Designer	Drag grip	Command exists	Reorders the command in the workflow.
BotJob Designer	Enabled checkbox	Command exists	Includes/excludes the command from execution; disabled API calls are excluded from Bash export.
BotJob Designer	Expand/collapse	Command exists	Shows or hides command configuration.
BotJob Designer	Command delete (x)	Command exists	Removes that command from the working definition.
Environments	+ New Environment	Always	Opens the environment form.
Environments	Star	Non-default custom environment	Sets the environment as default.
Environments	Edit	Custom environment	Opens the existing environment in the form.
Environments	Delete	Custom environment	Deletes after confirmation; built-in Mock cannot be deleted.
Environment form	Close / Cancel	Form open	Discards unsaved edits.
Environment form	Set as default	Form open	Makes the saved environment the default selection.
Environment form	Save	Name and Base URL present	Validates header lines and persists the environment.
Scripts	BotJob selector	Always	Selects the saved workflow used as the script source.
Scripts	Environment selector	Always	Selects base URL and default headers.
Scripts	Browse	Desktop app	Selects the exact output directory.
Scripts	Create Bash Script	Job, environment, and directory selected	Writes a timestamped .sh file containing enabled API_CALL curl commands.
Execute Tests	History / Hide History	Always	Shows or hides the saved-run list.
Execute Tests	Environment selector	Always	Chooses the run target and script handoff environment.
Execute Tests	BotJob selector	Always	Chooses the workflow; resets the displayed result.
Execute Tests	Create Scripts	BotJob selected	Opens Scripts with job and environment query selections.
Execute Tests	Run	BotJob selected	Creates a local run record and step audit.
Execute Tests	Step row	Result loaded	Expands errors, assertions, request, and response when recorded.
Execution History	All BotJobs filter	History open	Filters saved runs by BotJob.
Execution History	Run row	Runs exist	Loads that run and closes History.

Page	Control	Available when	Result
Mock Server	Start / Stop	Always	Changes mock dashboard state.
Mock Server	Clear	Log contains rows	Clears the in-app request/system log.
Reports	Execution Run selector	Runs exist	Chooses the source run for HTML/CSV.
Reports	HTML Download	Run selected	Downloads a human-readable run report.
Reports	CSV Download	Run selected	Downloads step results for spreadsheet/BI use.
Reports	Postman Download	Always	Downloads the whole catalog as Postman Collection v2.1 JSON.
Reports	Bash/curl Download	Always	Downloads a simple whole-catalog Bash file; review its localhost base URL.
Settings	Provider expand/collapse	Always	Shows or hides editable model/base URL/API-key fields.
Settings	Save	Provider expanded	Persists the provider record; blank key preserves an existing key.
Settings	Test	Saved key exists	Shows the current build's provider configuration test result.
Settings	SET DEFAULT	Saved key exists and provider is not default	Makes the provider active for AI features.

Appendix B. Local API reference

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Administrator/developer reference: These JSON routes are localhost-only implementation interfaces. Normal client operation should use the ARAPI UI and generated Bash files. Never publish this API publicly.

Method	Route	Purpose
GET	/health	Local backend liveness.
GET	/catalog/endpoints	List imported endpoints and mapping status.
POST	/import	Import a local folder from { folderPath }.
POST	/import/upload	Import uploaded OpenAPI/Swagger file content.
PUT	/catalog/endpoints/{endpointId}/category	Set or clear endpoint category mapping.
GET	/taxonomy	Return 25 banking categories and subcategories.
GET	/agents	List banking agents.
GET	/agents/capabilities	List agent capability counts.
POST	/agents/ask	Ask a banking agent.
POST	/app-assistant/chat	Send Bot Builder conversation messages.
GET	/settings/ai-providers	List provider settings without returning key values.
POST	/settings/ai-providers	Create or update a provider record.
POST	/settings/ai-providers/set-default	Set the default provider record.
POST	/settings/ai-providers/test	Return the current provider-test result.
GET	/environments	List environments.
POST	/environments	Create an environment.
PUT	/environments/{id}	Update an environment.
DELETE	/environments/{id}	Delete a non-built-in environment.
GET	/botjobs	List BotJobs.
POST	/botjobs	Create a BotJob shell.
GET	/botjobs/{id}	Get BotJob blocks, commands, and variables.
PUT	/botjobs/{id}	Save the complete BotJob definition.
DELETE	/botjobs/{id}	Delete a BotJob.
GET	/botjobs/{id}/export/bash	Generate BotJob-specific Bash; accepts environmentId query.
POST	/botjobs/{id}/execute	Create a run using { environmentId }.
GET	/executions	List runs; optional botJobId query filter.
GET	/executions/{runId}/steps	List step results for a run.

Method	Route	Purpose
GET	/executions/{runId}/report.html	Download HTML run report.
GET	/executions/{runId}/report.csv	Download CSV run report.
GET	/catalog/export/postman	Download whole-catalog Postman JSON; optional baseUrl query.
GET	/catalog/export/bash	Download whole-catalog Bash; optional baseUrl query.
GET	/mock/status	Read mock dashboard state and port.
POST	/mock/start	Set mock state to running and add a system log row.
POST	/mock/stop	Set mock state to stopped and add a system log row.
GET	/mock/log	Read the mock log.
POST	/mock/log/clear	Clear the mock log.
GET	/separation/progress	Read packaged progress metadata when present.

Appendix C. Glossary

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Term	Meaning
API Catalog	The imported endpoint inventory that constrains selectable API calls.
BotJob	A named ordered workflow containing commands and variables.
Command	One step in a BotJob, such as API_CALL or ASSERT_STATUS_CODE.
Environment	A named base URL plus optional default headers.
Mapping	Association between an endpoint and a banking business category.
Mock	A local/synthetic test context intended to avoid a production banking backend.
Run	One saved execution/audit record for a BotJob and environment.
Secret variable	A BotJob variable that Bash export requires from the process environment.
Synthetic endpoint	A mock-only catalog record that Bot Builder may create when no imported endpoint matches strongly enough.
Token	A \${name} placeholder resolved from BotJob/runtime variables.

Appendix D. Client acceptance checklist

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Application

- ARAPI version and distribution type are recorded.
- sidecar online appears after a clean launch.
- Approved backup/restore location is documented and tested.
- Local ports are not exposed publicly.

Catalog and workflow

- A known-good specification imports with expected endpoint count.
- A sample catalog endpoint is searchable and correctly classified or marked unmapped.
- A BotJob can be created, configured, reordered, and saved.
- Preview commands are excluded from acceptance criteria.

Bash execution

- The Scripts page saves to Documents\ARAPI\Scripts or an approved equivalent.
- The generated file contains only intended enabled API_CALL commands.
- Secrets appear as required environment variables, not literal values.
- A mock/dev script returns the expected process exit code and target-side evidence.

Audit and AI

- Execution History reloads a saved run and Reports downloads HTML/CSV.
- Whole-catalog Postman/Bash files are reviewed for localhost base URL before use.
- AI provider configuration is verified with a real Bot Builder request.
- Operators understand synthetic endpoint labels and do not treat them as imported production APIs.

Sign-off: Client acceptance should record approver, date, environment, BotJob/script identifier, evidence location, exceptions, and remediation owner.

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